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Lucas et al.

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(54) **SELECTING A TILT ANGLE OF AN AR DISPLAY**

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(52) **U.S. Cl.**

CPC **G06T 19/006** (2013.01); **G02B 27/0172** (2013.01); **G06F 3/013** (2013.01); **G06T 19/20** (2013.01); **G02B 2027/0178** (2013.01)

(58) **Field of Classification Search**

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See application file for complete search history.

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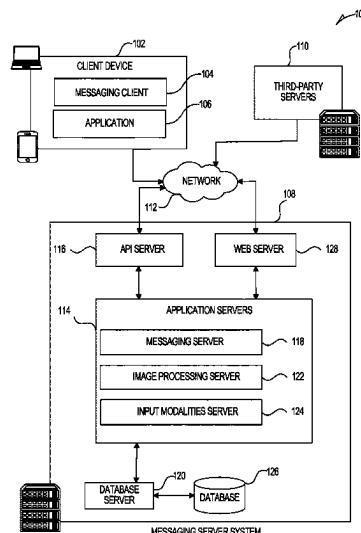
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ABSTRACT

Systems, methods, and computer readable media for selecting a tilt angle of an augmented reality (AR) display of an AR wearable device. Some examples of the present disclosure capture simulation data of gaze fixations while users are performing tasks using applications resident on the AR wearable device. The tilt angle of the AR display is selected based on including more gaze fixations that are within the field of view (FOV) of the AR display than are outside the FOV of the AR display. In some examples, an AR wearable device is manufactured with a fixed vertical tilt angle for the AR display. In some examples, the AR wearable device can dynamically adjust the vertical tilt angle of the AR display based on the applications that a user of the AR wearable device is likely to use or is using.

20 Claims, 16 Drawing Sheets



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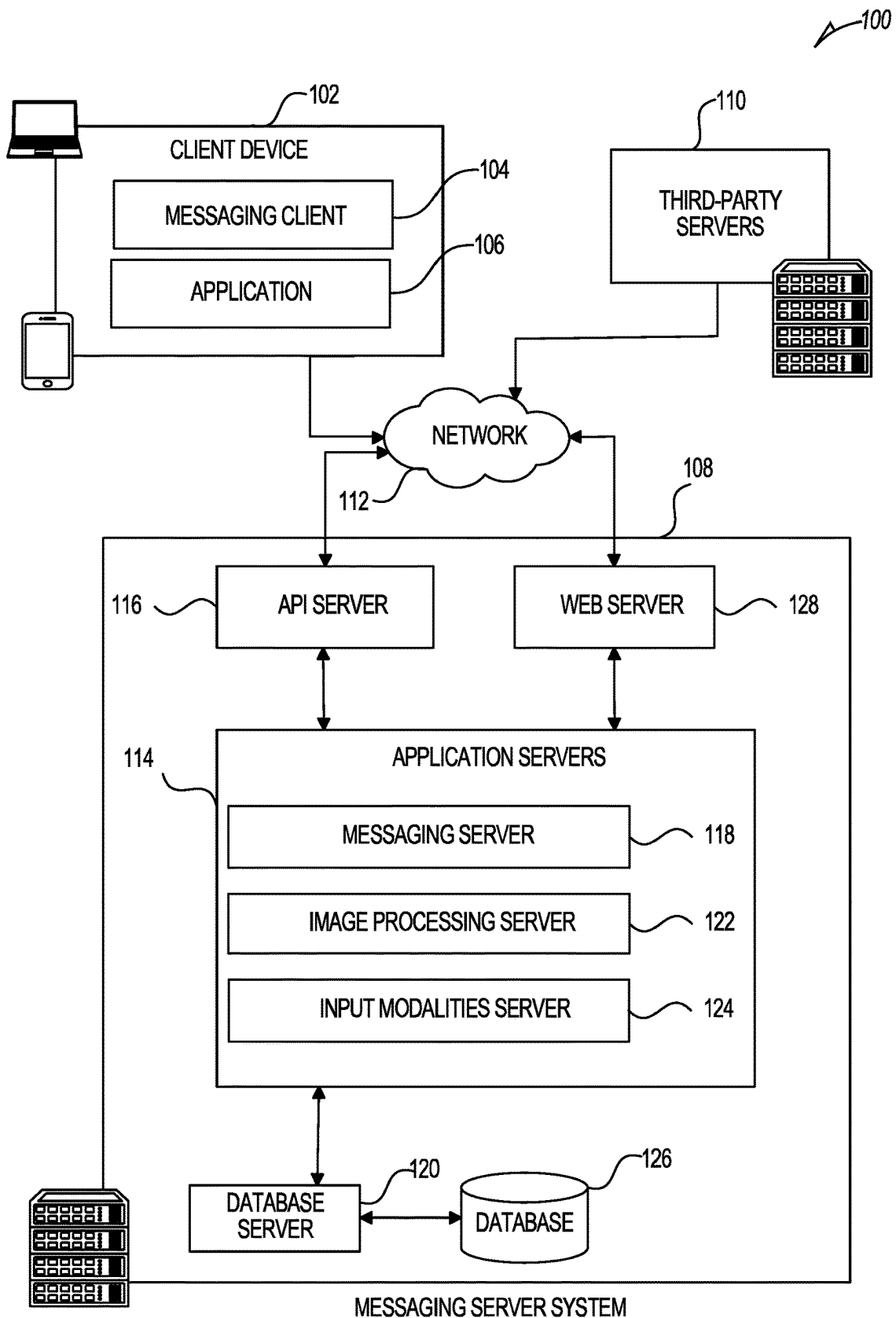


FIG. 1

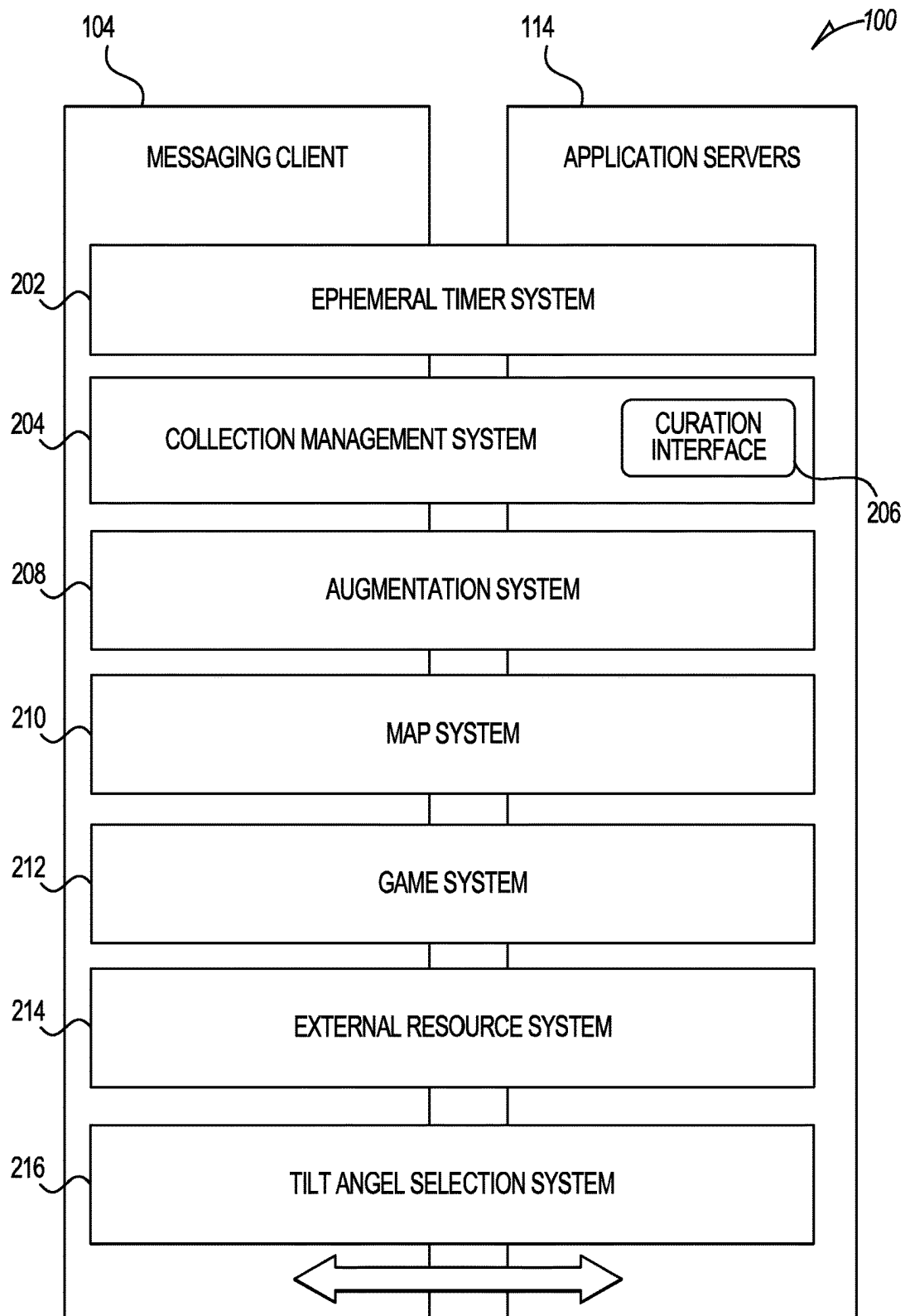


FIG. 2

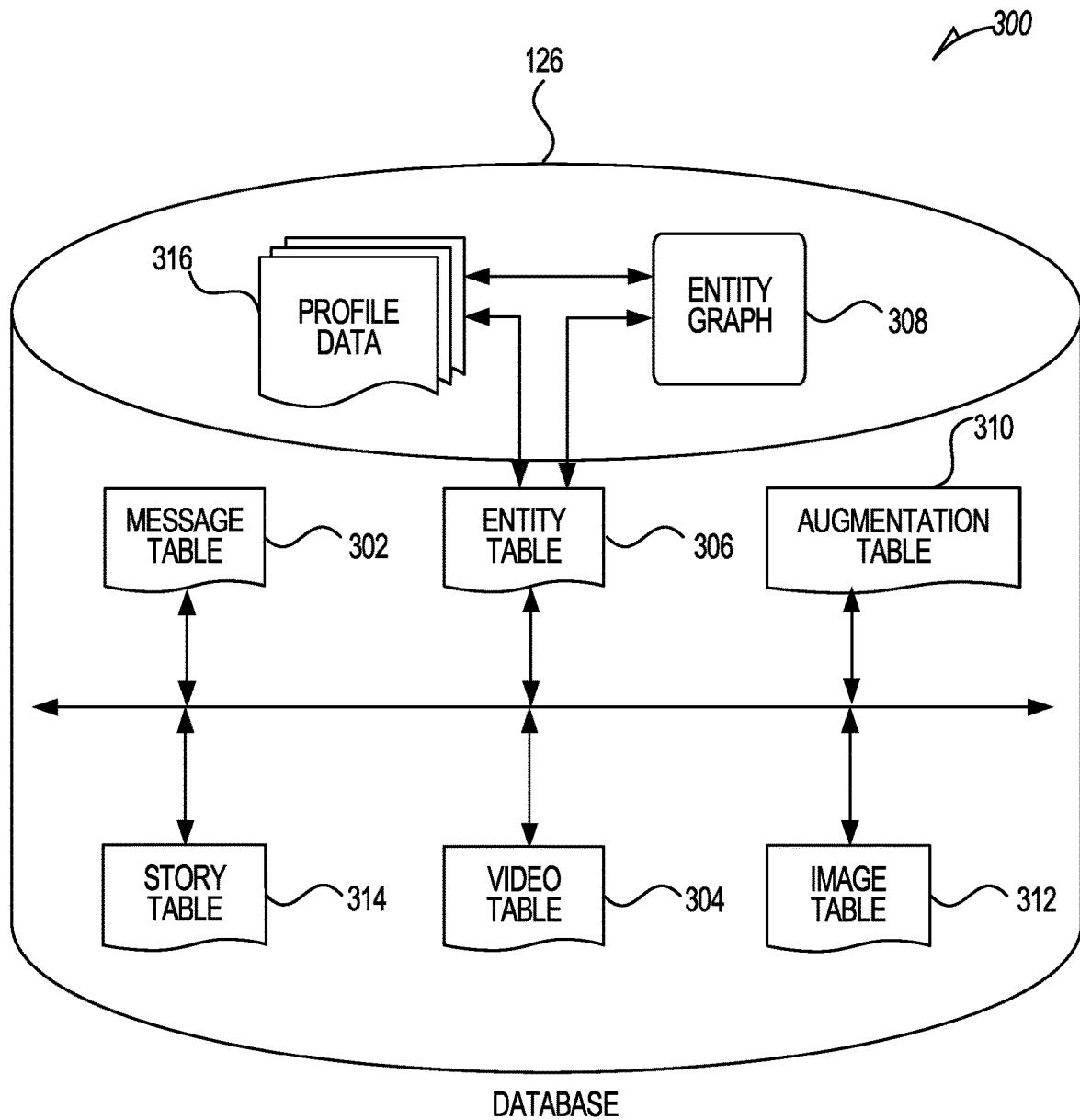


FIG. 3

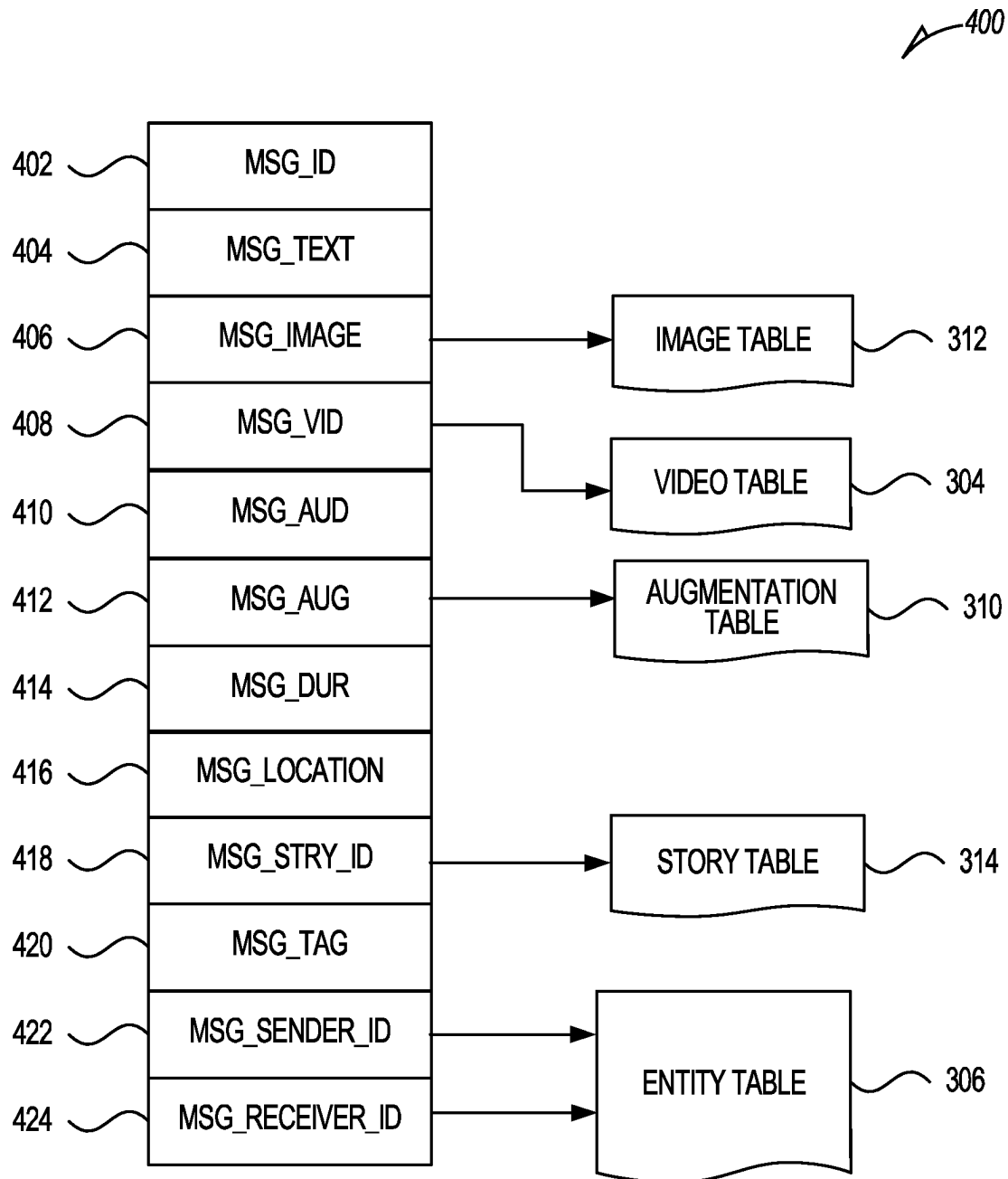


FIG. 4

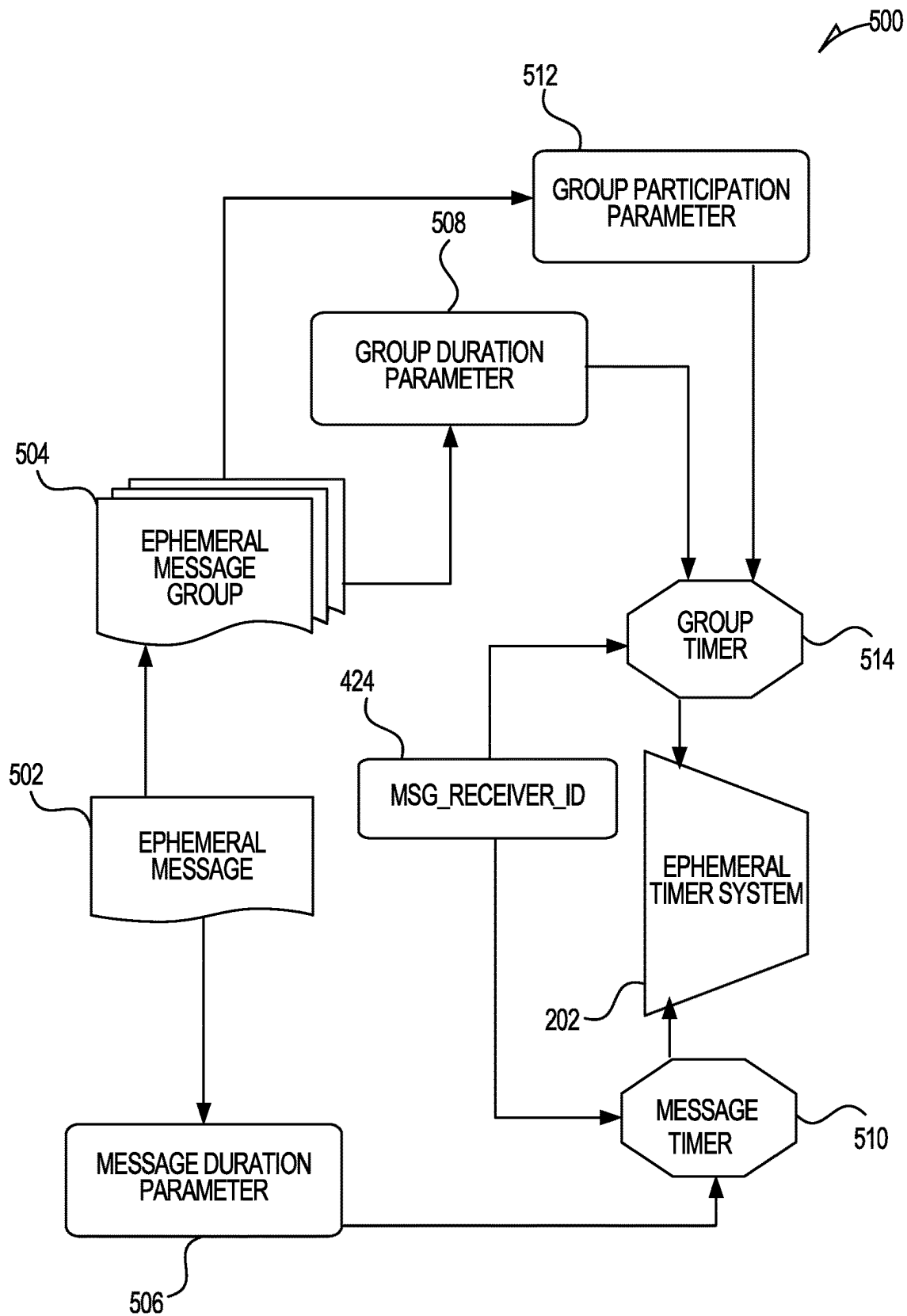


FIG. 5

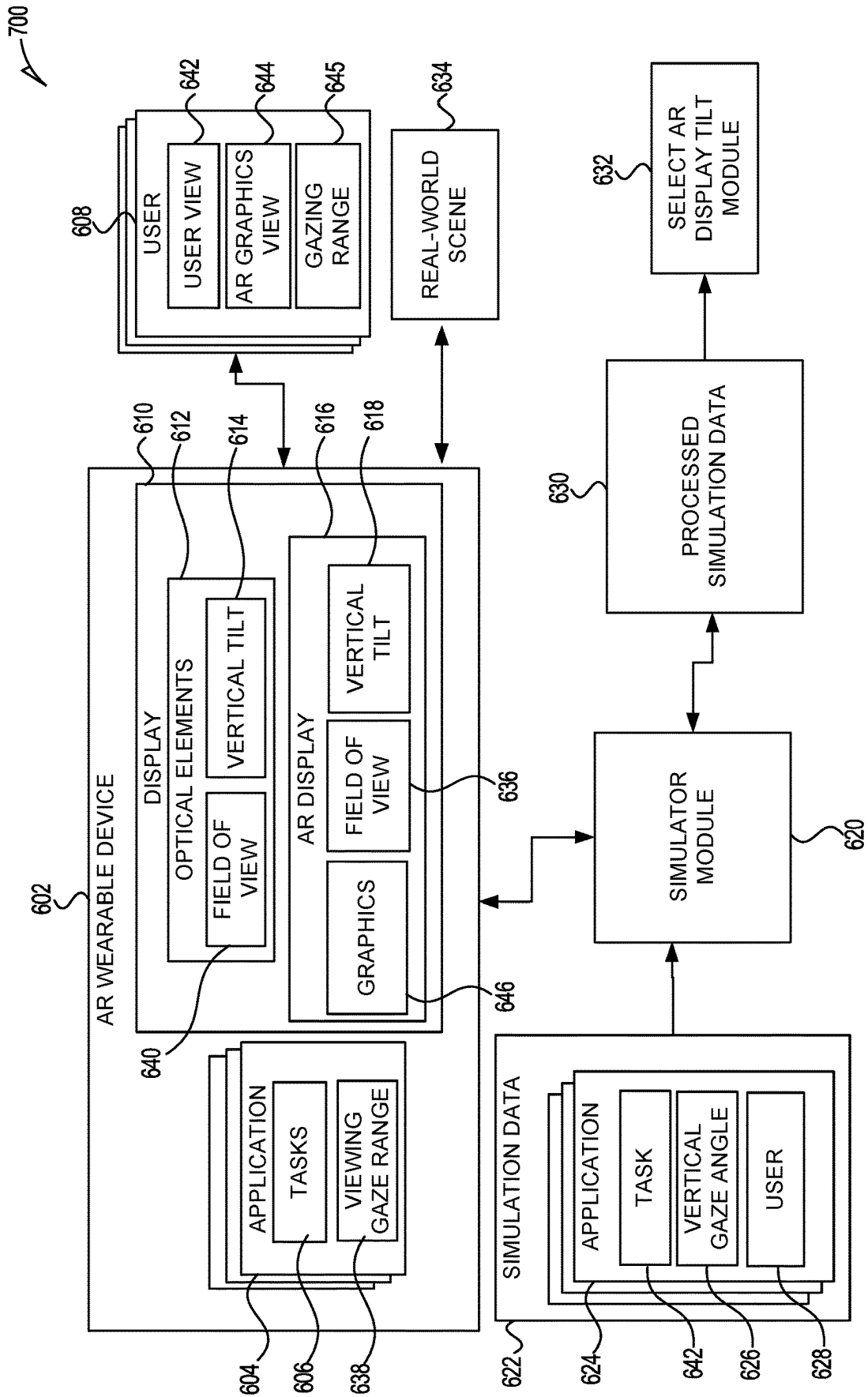


FIG. 6

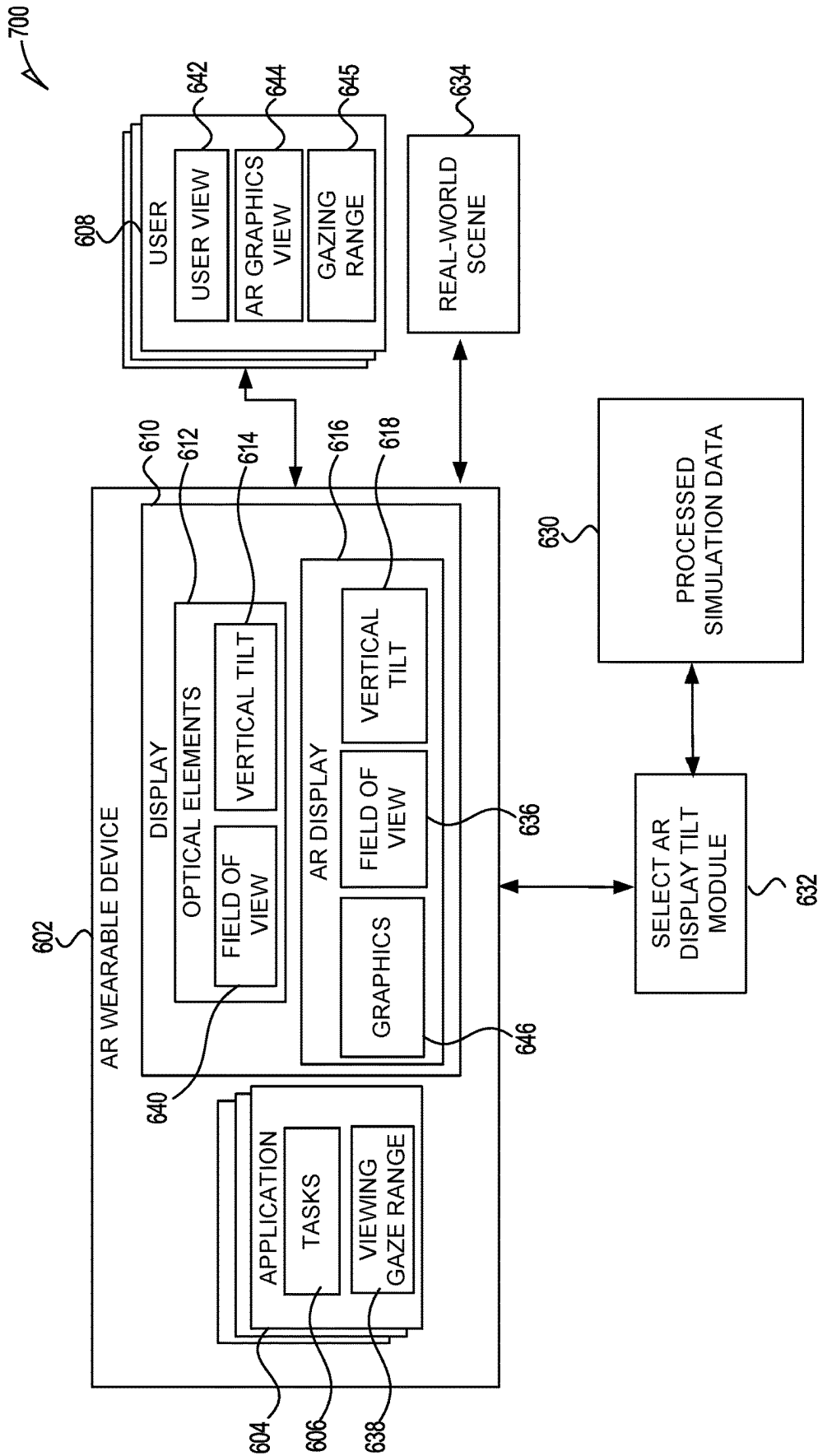


FIG. 7

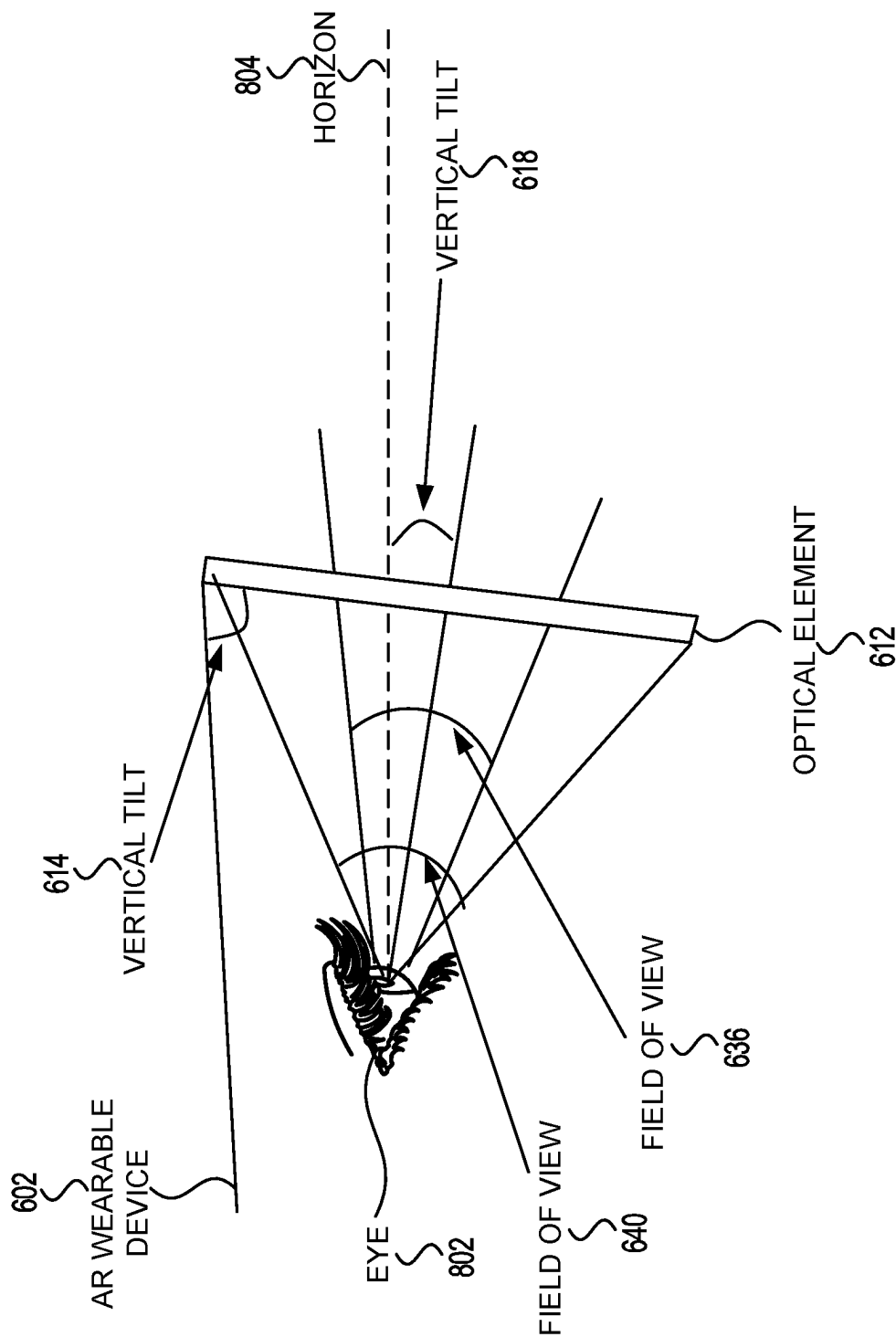


FIG. 8

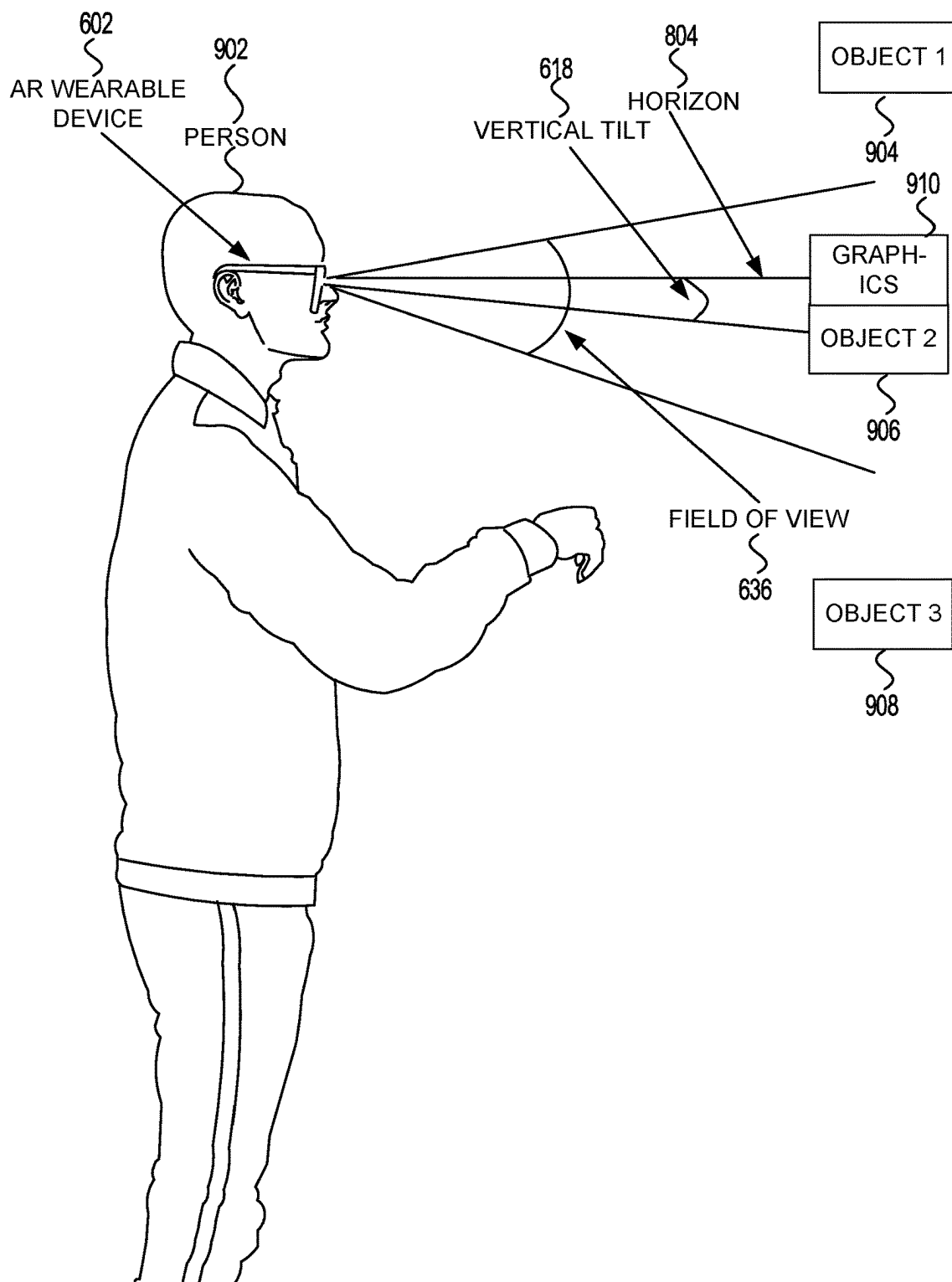


FIG. 9

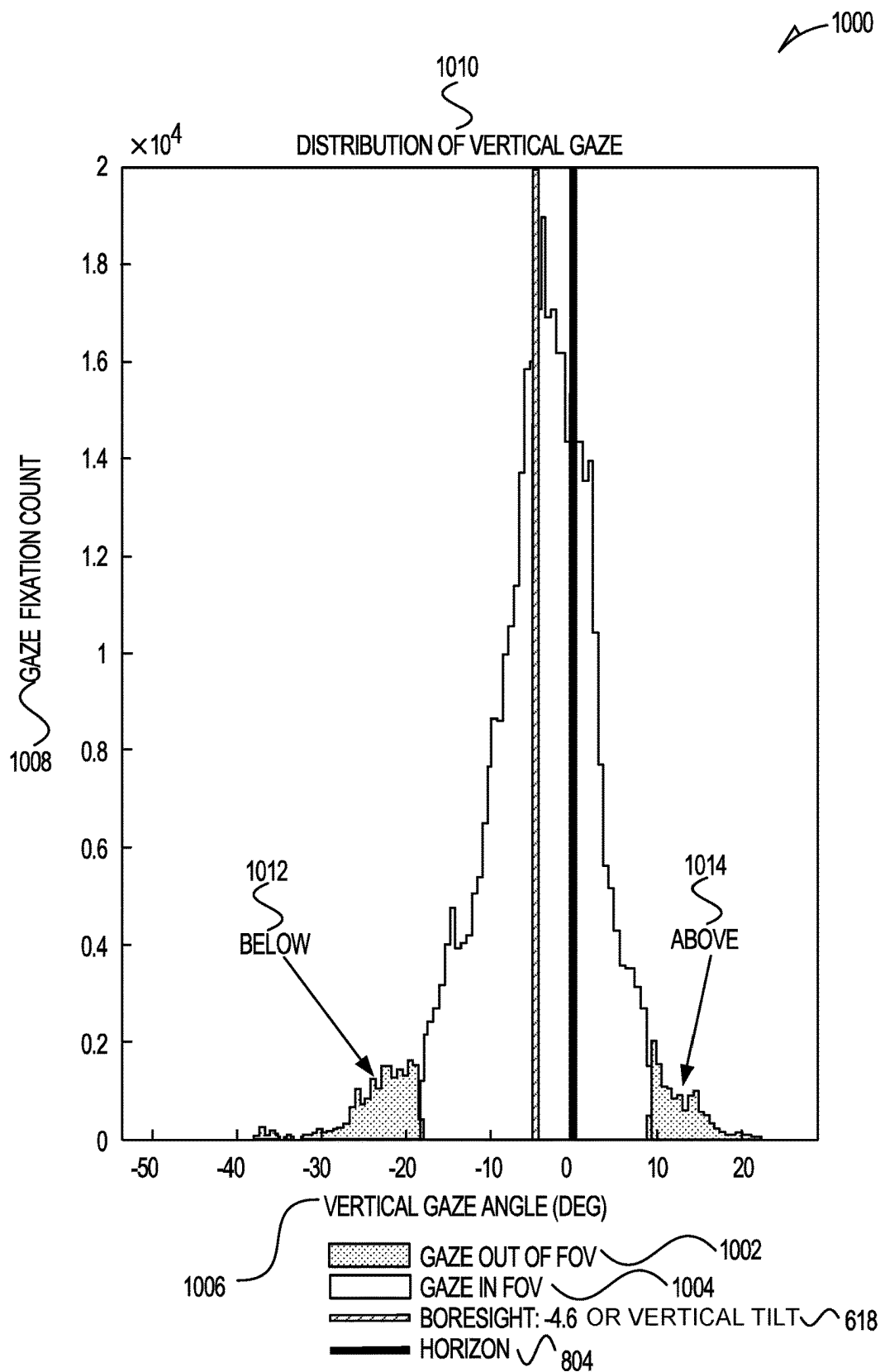


FIG. 10

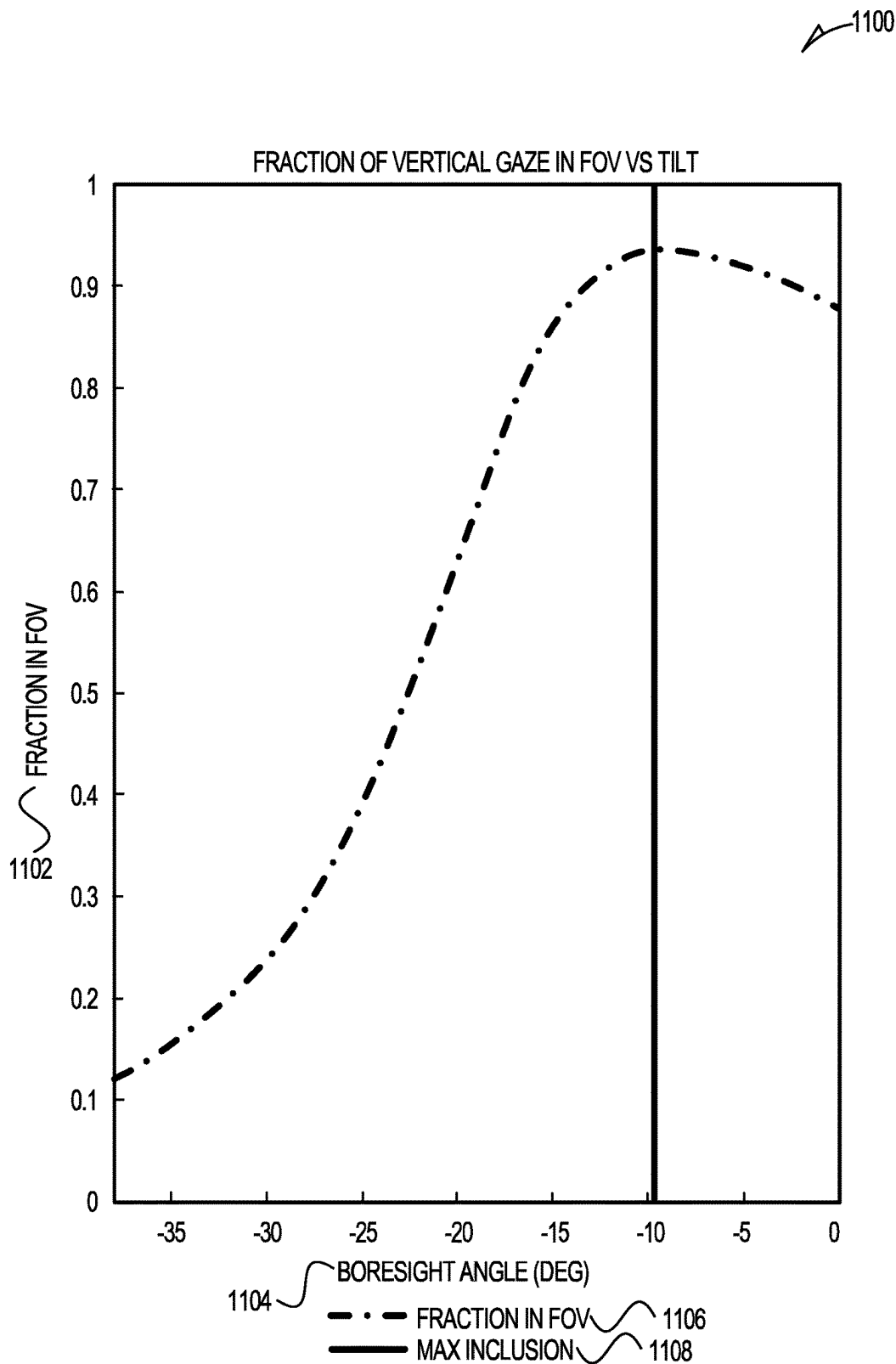


FIG. 11

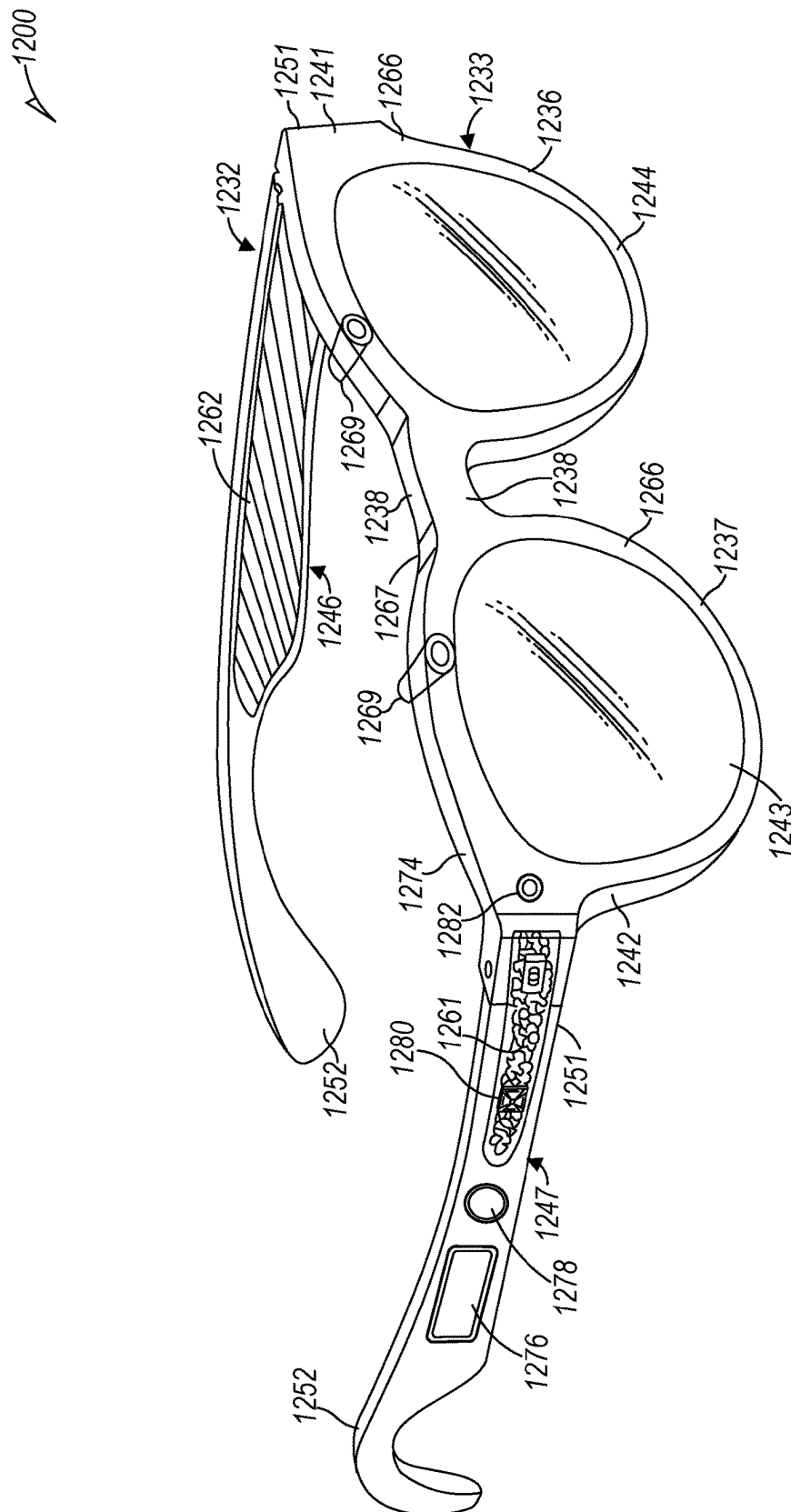


FIG. 12

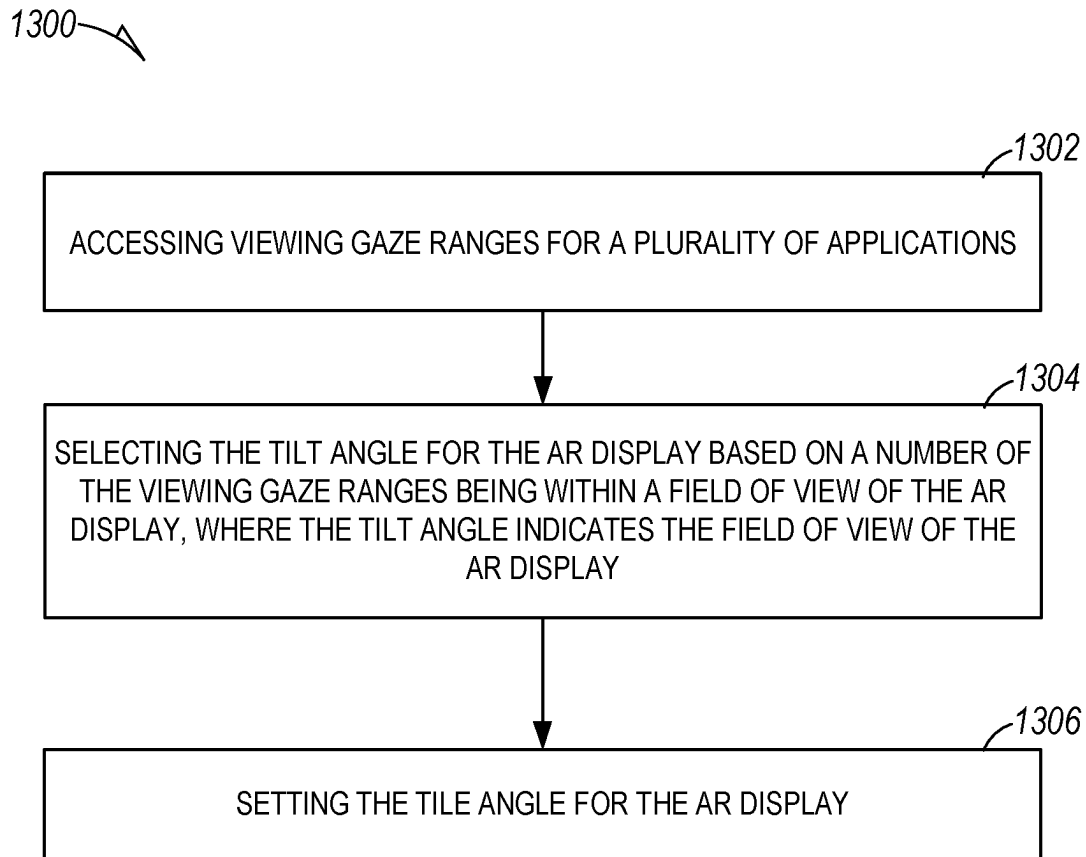


FIG. 13

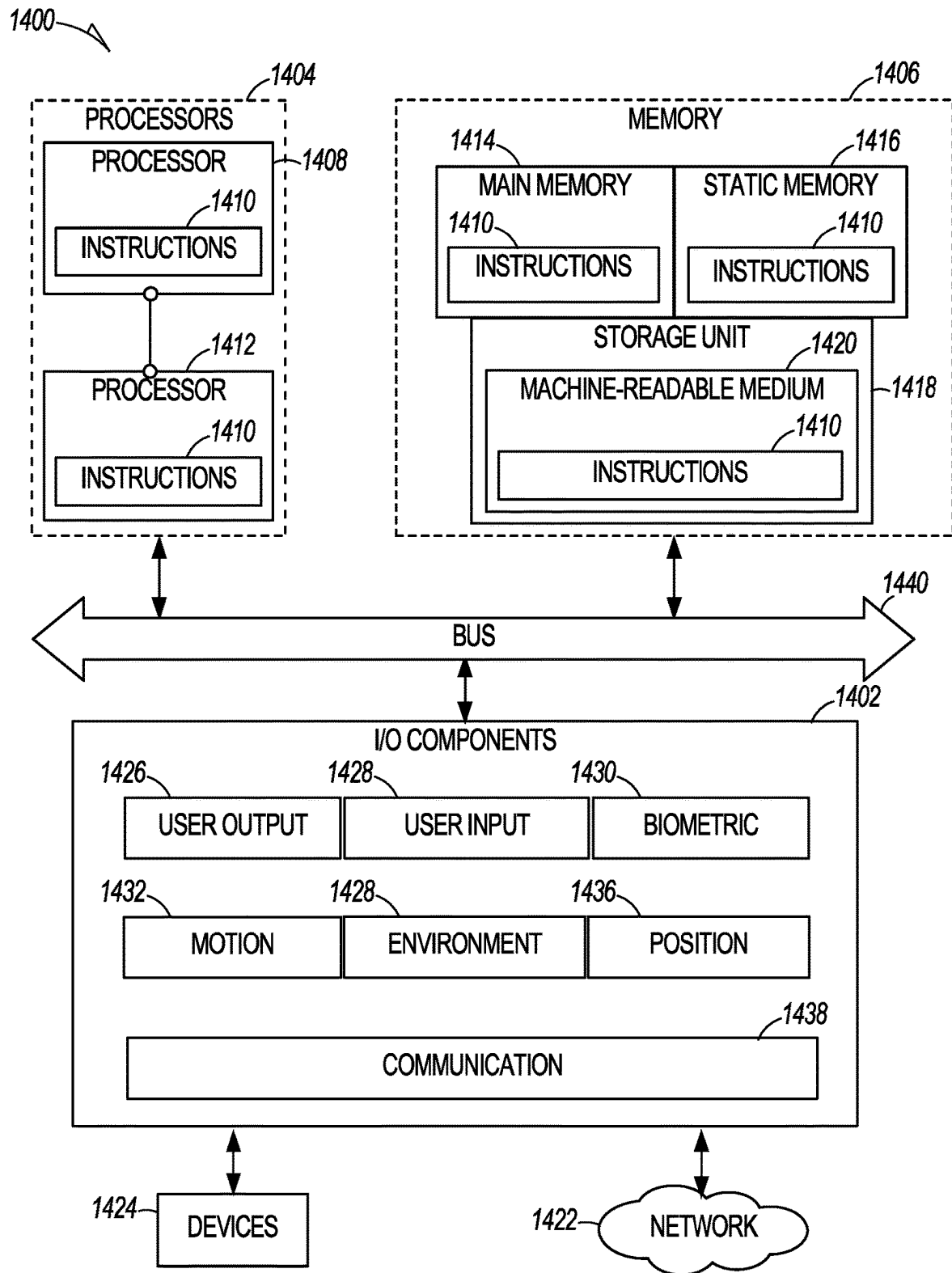


FIG. 14

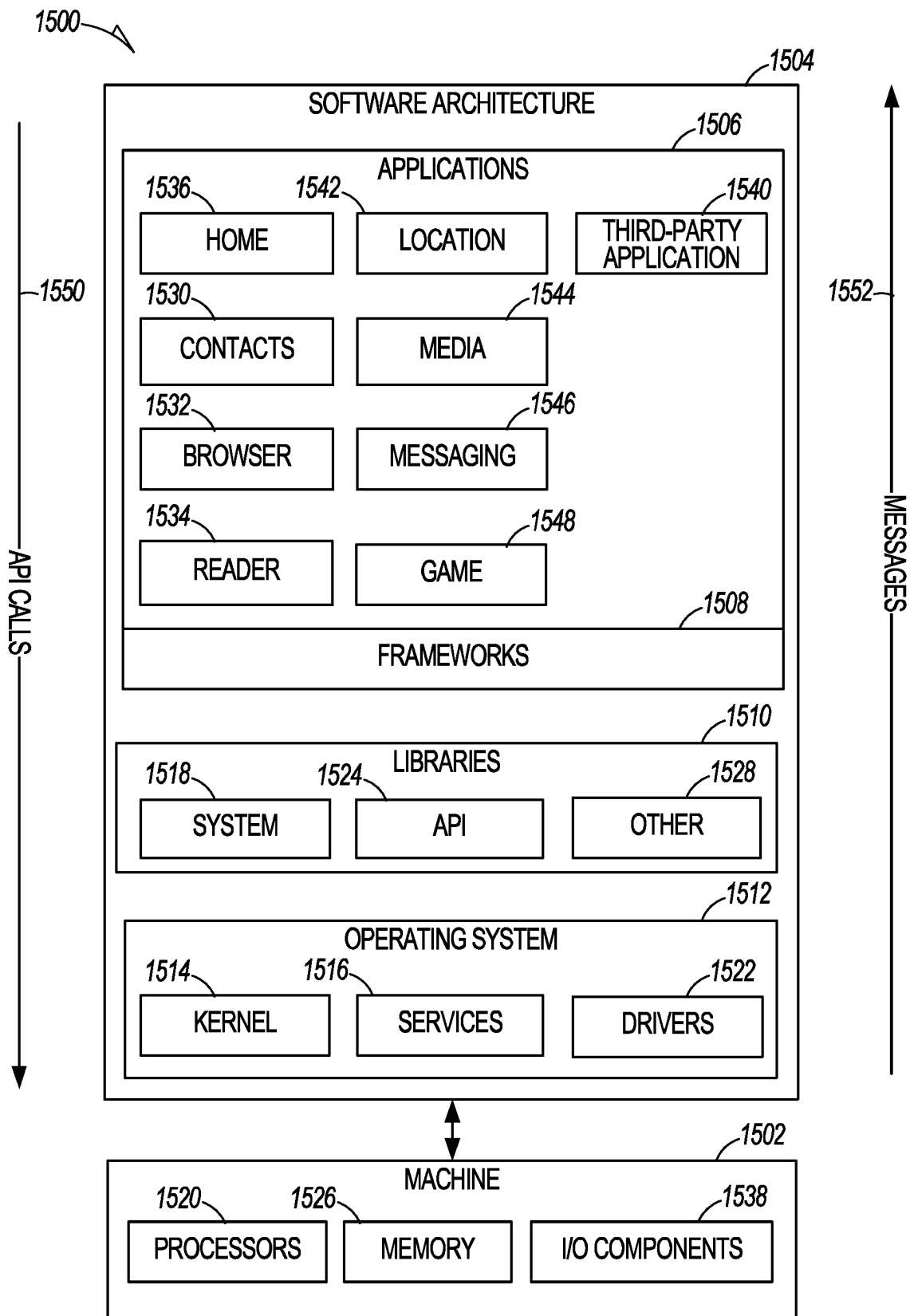


FIG. 15

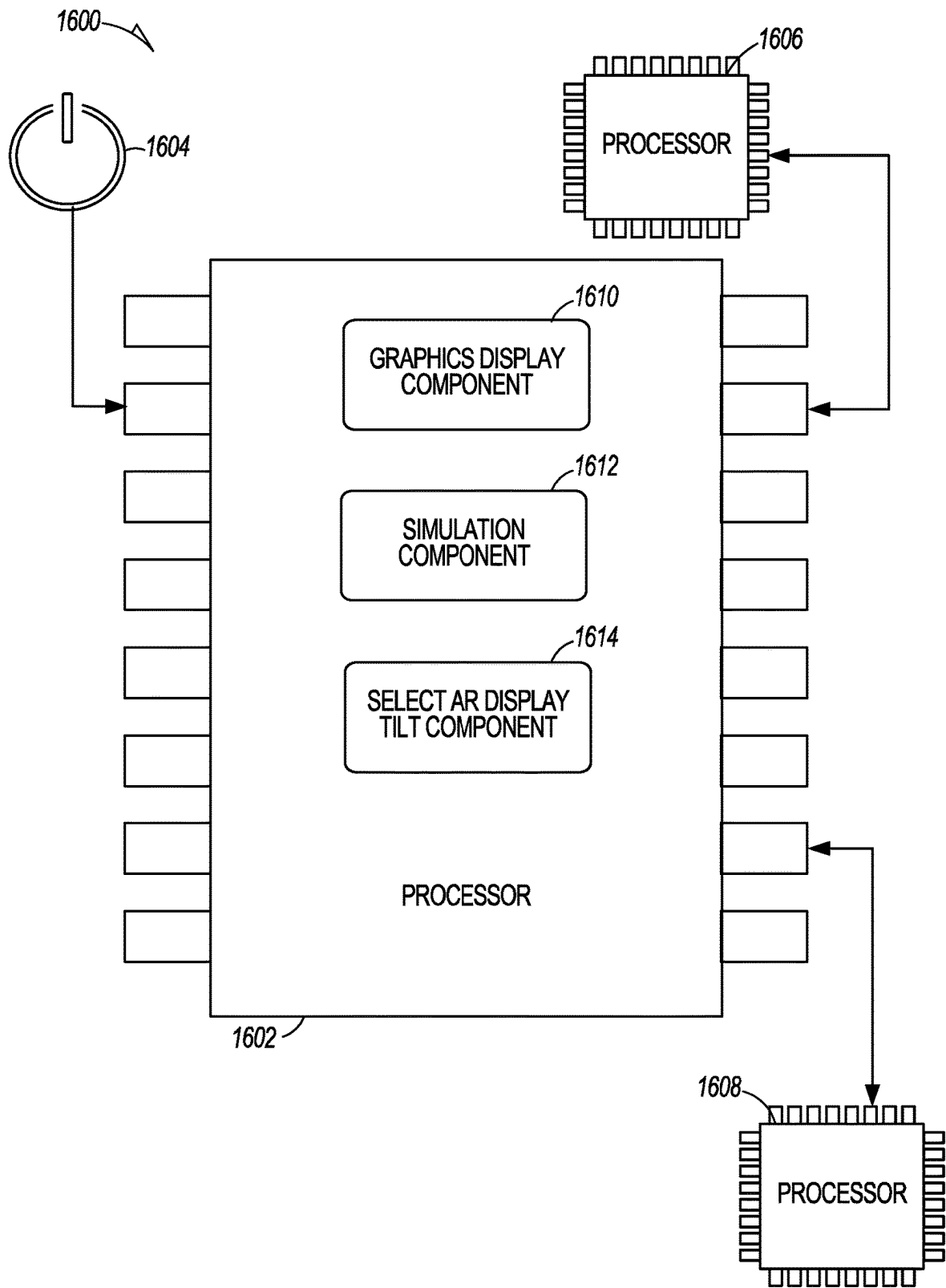


FIG. 16

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SELECTING A TILT ANGLE OF AN AR DISPLAY

TECHNICAL FIELD

Examples of the present disclosure relate generally to selecting a tilt angle of an augmented reality (AR) display of an AR wearable device. More particularly, but not by way of limitation, examples of the present disclosure relate to capturing simulation data of gaze fixations and selecting a tilt angle that accommodates gaze fixations for applications and tasks performed by users of the AR wearable devices.

BACKGROUND

Users increasingly want virtual reality (VR), mixed reality (MR), and augmented reality (AR) wearable devices to operate in a more user-friendly manner with more functions. However, often, the wearable devices have limited displays and often the AR wearable devices have limited power to provide additional functions.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

In the drawings, which are not necessarily drawn to scale, like numerals may describe similar components in different views. To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced. Some nonlimiting examples are illustrated in the figures of the accompanying drawings in which:

FIG. 1 is a diagrammatic representation of a networked environment in which the present disclosure may be deployed, in accordance with some examples.

FIG. 2 is a diagrammatic representation of a messaging system, in accordance with some examples, that has both client-side and server-side functionality.

FIG. 3 is a diagrammatic representation of a data structure as maintained in a database, in accordance with some examples.

FIG. 4 is a diagrammatic representation of a message, in accordance with some examples.

FIG. 5 is a flowchart for an access-limiting process, in accordance with some examples.

FIG. 6 illustrates a system for selecting a tilt angle of an AR display, in accordance with some examples.

FIG. 7 illustrates a system for selecting a tilt angle of an AR display, in accordance with some examples.

FIG. 8 illustrates the vertical tilt angle, in accordance with some examples.

FIG. 9 illustrates the vertical tilt angle, in accordance with some examples.

FIG. 10 illustrates a graph, in accordance with some examples.

FIG. 11 illustrates a graph, in accordance with some examples.

FIG. 12 is a perspective view of a wearable electronic device in the form of glasses 1200, in accordance with some examples.

FIG. 13 illustrates a method for selecting a tilt angle of an AR display, in accordance with some examples.

FIG. 14 is a diagrammatic representation of a machine in the form of a computer system within which a set of instructions may be executed for causing the machine to

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perform any one or more of the methodologies discussed herein, in accordance with some examples.

FIG. 15 is a block diagram showing a software architecture within which examples may be implemented.

FIG. 16 is a diagrammatic representation of a processing environment, in accordance with some examples.

DETAILED DESCRIPTION

The description that follows includes systems, methods, techniques, instruction sequences, and computing machine program products that embody illustrative examples of the disclosure. In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide an understanding of various examples of the inventive subject matter. It will be evident, however, to those skilled in the art, that examples of the inventive subject matter may be practiced without these specific details. In general, well-known instruction instances, protocols, structures, and techniques are not necessarily shown in detail.

The term AR wearable device is used as an illustrative device; however, one skilled in the art will recognize that the methods, systems, and computer readable medium disclosed herein are applicable to other wearable devices including VR wearable device and MR wearable devices.

AR wearable devices 602 such as AR glasses 1200 have a limited field of view (FOV) for displaying graphics or AR graphics on the AR display 616. The limited FOV is less than the full FOV of a user 608. For example, a user 608 may have a gaze fixation of 35 degree by gazing upwards without moving the AR wearable device 602 they are wearing. The gaze fixation may be on an object onto which the AR display 616 is not able to project graphics.

One challenge is selecting a vertical tilt 618 angle of the AR display 616 relative to a horizon 804 to lessen the number of gaze fixations of a user 608 that fall outside a FOV of the AR display 616 and increase the number of gaze fixations of a user 608 that fall inside a FOV of the AR display 616. For example, referring to FIG. 9, the AR display 616 can display graphics 910 to enhance object 2 906 but cannot generate graphics to enhance object 1 904 or object 3 908 because object 1 904 and object 3 908 are outside the FOV of the AR display 616 but object 2 906 is inside the FOV of the AR display 616.

In some examples, the challenge is addressed by collecting gaze fixation data from users 608 while they are performing tasks 606 using applications 604 such as selecting a breakfast cereal from a food market. The gaze fixation data is collected and then processed to generate a probability distribution of vertical viewing angles or different processed data that can be used to determine a vertical tilt 618 angle for the AR display 616 that increases or maximizes the gaze fixations of users 608 within the FOV of the AR display 616 while the user 608 is performing tasks 606 using applications 604. The tasks 606 using applications 604 are selected based on a likelihood the user 608 will use perform the tasks 606 based on previous behavior of the user 608 or previous behavior of many users 608 using the AR wearable device 602. In some examples, the AR wearable devices 602 are manufactured with a fixed vertical tilt 618 angle and in some examples, the AR wearable device 602 can adjust the vertical tilt 618 angle in accordance with a set of tasks 606 the user 608 is likely to perform or is performing.

Networked Computing Environment

FIG. 1 is a block diagram showing an example messaging system 100 for exchanging data (e.g., messages and associated content) over a network. The messaging system 100

includes multiple instances of a client device **102**, each of which hosts a number of applications, including a messaging client **104** and other applications **106**. Each messaging client **104** is communicatively coupled to other instances of the messaging client **104** (e.g., hosted on respective other client devices **102**), a messaging server system **108** and third-party servers **110** via a network **112** (e.g., the Internet). A messaging client **104** can also communicate with locally-hosted applications **106** using Applications Program Interfaces (APIs).

A messaging client **104** is able to communicate and exchange data with other messaging clients **104** and with the messaging server system **108** via the network **112**. The data exchanged between messaging clients **104**, and between a messaging client **104** and the messaging server system **108**, includes functions (e.g., commands to invoke functions) as well as payload data (e.g., text, audio, video or other multimedia data).

The messaging server system **108** provides server-side functionality via the network **112** to a particular messaging client **104**. While certain functions of the messaging system **100** are described herein as being performed by either a messaging client **104** or by the messaging server system **108**, the location of certain functionality either within the messaging client **104** or the messaging server system **108** may be a design choice. For example, it may be technically preferable to initially deploy certain technology and functionality within the messaging server system **108** but to later migrate this technology and functionality to the messaging client **104** where a client device **102** has sufficient processing capacity.

The messaging server system **108** supports various services and operations that are provided to the messaging client **104**. Such operations include transmitting data to, receiving data from, and processing data generated by the messaging client **104**. This data may include message content, client device information, geolocation information, media augmentation and overlays, message content persistence conditions, social network information, and live event information, as examples. Data exchanges within the messaging system **100** are invoked and controlled through functions available via user interfaces (UIs) of the messaging client **104**.

Turning now specifically to the messaging server system **108**, an Application Program Interface (API) server **116** is coupled to, and provides a programmatic interface to, application servers **114**. The application servers **114** are communicatively coupled to a database server **120**, which facilitates access to a database **126** that stores data associated with messages processed by the application servers **114**. Similarly, a web server **128** is coupled to the application servers **114**, and provides web-based interfaces to the application servers **114**. To this end, the web server **128** processes incoming network requests over the Hypertext Transfer Protocol (HTTP) and several other related protocols.

The Application Program Interface (API) server **116** receives and transmits message data (e.g., commands and message payloads) between the client device **102** and the application servers **114**. Specifically, the Application Program Interface (API) server **116** provides a set of interfaces (e.g., routines and protocols) that can be called or queried by the messaging client **104** in order to invoke functionality of the application servers **114**. The Application Program Interface (API) server **116** exposes various functions supported by the application servers **114**, including account registration, login functionality, the sending of messages, via the application servers **114**, from a particular messaging client

104 to another messaging client **104**, the sending of media files (e.g., images or video) from a messaging client **104** to a messaging server **118**, and for possible access by another messaging client **104**, the settings of a collection of media data (e.g., story), the retrieval of a list of friends of a user of a client device **102**, the retrieval of such collections, the retrieval of messages and content, the addition and deletion of entities (e.g., friends) to an entity graph (e.g., a social graph), the location of friends within a social graph, and opening an application event (e.g., relating to the messaging client **104**).

The application servers **114** host a number of server applications and subsystems, including for example a messaging server **118**, an image processing server **122**, and an input modalities server **124**. The messaging server **118** implements a number of message processing technologies and functions, particularly related to the aggregation and other processing of content (e.g., textual and multimedia content) included in messages received from multiple instances of the messaging client **104**. As will be described in further detail, the text and media content from multiple sources may be aggregated into collections of content (e.g., called stories or galleries). These collections are then made available to the messaging client **104**. Other processor and memory intensive processing of data may also be performed server-side by the messaging server **118**, in view of the hardware requirements for such processing.

The application servers **114** also include an image processing server **122** that is dedicated to performing various image processing operations, typically with respect to images or video within the payload of a message sent from or received at the messaging server **118**.

The input modalities server **124** supports input modalities for AR wearable devices. The input modalities server **124** receives requests from an AR wearable device and responds to the requests. The requests include sensor data such as an image being sent to the input modalities server **124** for processing. The input modalities server **124** processes the sensor data and identifies objects within the sensor data and returns names of the objects and positions of the objects within the sensor data to the AR wearable device. Another request from the AR wearable device is for AR applications associated with tags such as "QR code" that may be run on the AR wearable device. The input modalities server **124** may load the AR wearable device with AR applications that are likely to be used by a user of the AR wearable device or respond with AR applications based on criteria given to the input modalities server **124** from the AR wearable device. The criteria may be as a limit on the number of AR applications, preferences of the user such as AR applications with links back to the messaging system **100**, and so forth.

Returning to the messaging client **104**, features and functions of an external resource (e.g., an application **106** or applet) are made available to a user via an interface of the messaging client **104**. In this context, "external" refers to the fact that the application **106** or applet is external to the messaging client **104**. The external resource is often provided by a third party but may also be provided by the creator or provider of the messaging client **104**. The messaging client **104** receives a user selection of an option to launch or access features of such an external resource. The external resource may be the application **106** installed on the client device **102** (e.g., a "native app"), or a small-scale version of the application (e.g., an "applet") that is hosted on the client device **102** or remote of the client device **102** (e.g., on third-party servers **110**). The small-scale version of the application includes a subset of features and functions of the

application (e.g., the full-scale, native version of the application) and is implemented using a markup-language document. In one example, the small-scale version of the application (e.g., an “applet”) is a web-based, markup-language version of the application and is embedded in the messaging client **104**. In addition to using markup-language documents (e.g., a *.ml file), an applet may incorporate a scripting language (e.g., a *.js file or a *.json file) and a style sheet (e.g., a *.ss file).

In response to receiving a user selection of the option to launch or access features of the external resource, the messaging client **104** determines whether the selected external resource is a web-based external resource or a locally-installed application **106**. In some cases, applications **106** that are locally installed on the client device **102** can be launched independently of and separately from the messaging client **104**, such as by selecting an icon, corresponding to the application **106**, on a home screen of the client device **102**. Small-scale versions of such applications can be launched or accessed via the messaging client **104** and, in some examples, no or limited portions of the small-scale application can be accessed outside of the messaging client **104**. The small-scale application can be launched by the messaging client **104** receiving, from a third-party server **110** for example, a markup-language document associated with the small-scale application and processing such a document.

In response to determining that the external resource is a locally-installed application **106**, the messaging client **104** instructs the client device **102** to launch the external resource by executing locally-stored code corresponding to the external resource. In response to determining that the external resource is a web-based resource, the messaging client **104** communicates with the third-party servers **110** (for example) to obtain a markup-language document corresponding to the selected external resource. The messaging client **104** then processes the obtained markup-language document to present the web-based external resource within a user interface of the messaging client **104**.

The messaging client **104** can notify a user of the client device **102**, or other users related to such a user (e.g., “friends”), of activity taking place in one or more external resources. For example, the messaging client **104** can provide participants in a conversation (e.g., a chat session) in the messaging client **104** with notifications relating to the current or recent use of an external resource by one or more members of a group of users. One or more users can be invited to join in an active external resource or to launch a recently-used but currently inactive (in the group of friends) external resource. The external resource can provide participants in a conversation, each using respective messaging clients **104**, with the ability to share an item, status, state, or location in an external resource with one or more members of a group of users into a chat session. The shared item may be an interactive chat card with which members of the chat can interact, for example, to launch the corresponding external resource, view specific information within the external resource, or take the member of the chat to a specific location or state within the external resource. Within a given external resource, response messages can be sent to users on the messaging client **104**. The external resource can selectively include different media items in the responses, based on a current context of the external resource.

The messaging client **104** can present a list of the available external resources (e.g., applications **106** or applets) to a user to launch or access a given external resource. This list can be presented in a context-sensitive menu. For example,

the icons representing different ones of the application **106** (or applets) can vary based on how the menu is launched by the user (e.g., from a conversation interface or from a non-conversation interface).

5 System Architecture

FIG. **2** is a block diagram illustrating further details regarding the messaging system **100**, according to some examples. Specifically, the messaging system **100** is shown to comprise the messaging client **104** and the application servers **114**. The messaging system **100** embodies a number of subsystems, which are supported on the client-side by the messaging client **104** and on the server-side by the application servers **114**. These subsystems include, for example, an ephemeral timer system **202**, a collection management system **204**, an augmentation system **208**, a map system **210**, a game system **212**, an external resource system **214**, and a tilt angle selection system **216**.

The ephemeral timer system **202** is responsible for enforcing the temporary or time-limited access to content by the messaging client **104** and the messaging server **118**. The ephemeral timer system **202** incorporates a number of timers that, based on duration and display parameters associated with a message, or collection of messages (e.g., a story), selectively enable access (e.g., for presentation and display) to messages and associated content via the messaging client **104**. Further details regarding the operation of the ephemeral timer system **202** are provided below.

The collection management system **204** is responsible for managing sets or collections of media (e.g., collections of text, image video, and audio data). A collection of content (e.g., messages, including images, video, text, and audio) may be organized into an “event gallery” or an “event story.” Such a collection may be made available for a specified time period, such as the duration of an event to which the content relates. For example, content relating to a music concert may be made available as a “story” for the duration of that music concert. The collection management system **204** may also be responsible for publishing an icon that provides notification of the existence of a particular collection to the user interface of the messaging client **104**.

The collection management system **204** furthermore includes a curation interface **206** that allows a collection manager to manage and curate a particular collection of content. For example, the curation interface **206** enables an event organizer to curate a collection of content relating to a specific event (e.g., delete inappropriate content or redundant messages). Additionally, the collection management system **204** employs machine vision (or image recognition technology) and content rules to automatically curate a content collection. In certain examples, compensation may be paid to a user for the inclusion of user-generated content into a collection. In such cases, the collection management system **204** operates to automatically make payments to such users for the use of their content.

The augmentation system **208** provides various functions that enable a user to augment (e.g., annotate or otherwise modify or edit) media content associated with a message. For example, the augmentation system **208** provides functions related to the generation and publishing of media overlays for messages processed by the messaging system **100**. The augmentation system **208** operatively supplies a media overlay or augmentation (e.g., an image filter) to the messaging client **104** based on a geolocation of the client device **102**. In another example, the augmentation system **208** operatively supplies a media overlay to the messaging client **104** based on other information, such as social network information of the user of the client device **102**. A

media overlay may include audio and visual content and visual effects. Examples of audio and visual content include pictures, texts, logos, animations, and sound effects. An example of a visual effect includes color overlaying. The audio and visual content or the visual effects can be applied to a media content item (e.g., a photo, a digital object) at the client device **102**. For example, the media overlay may include text or image that can be overlaid on top of a photograph taken by the client device **102**. In another example, the media overlay includes an identification of a location overlay (e.g., Venice beach), a name of a live event, or a name of a merchant overlay (e.g., Beach Coffee House). In another example, the augmentation system **208** uses the geolocation of the client device **102** to identify a media overlay that includes the name of a merchant at the geolocation of the client device **102**. The media overlay may include other indicia associated with the merchant. The media overlays may be stored in the database **126** and accessed through the database server **120**.

In some examples, the augmentation system **208** provides a user-based publication platform that enables users to select a geolocation on a map and upload content associated with the selected geolocation. The user may also specify circumstances under which a particular media overlay should be offered to other users. The augmentation system **208** generates a media overlay that includes the uploaded content and associates the uploaded content with the selected geolocation.

In other examples, the augmentation system **208** provides a merchant-based publication platform that enables merchants to select a particular media overlay associated with a geolocation via a bidding process. For example, the augmentation system **208** associates the media overlay of the highest bidding merchant with a corresponding geolocation for a predefined amount of time.

The map system **210** provides various geographic location functions and supports the presentation of map-based media content and messages by the messaging client **104**. For example, the map system **210** enables the display of user icons or avatars (e.g., stored in profile data **316**) on a map to indicate a current or past location of “friends” of a user, as well as media content (e.g., collections of messages including photographs and videos) generated by such friends, within the context of a map. For example, a message posted by a user to the messaging system **100** from a specific geographic location may be displayed within the context of a map at that particular location to “friends” of a specific user on a map interface of the messaging client **104**. A user can furthermore share his or her location and status information (e.g., using an appropriate status avatar) with other users of the messaging system **100** via the messaging client **104**, with this location and status information being similarly displayed within the context of a map interface of the messaging client **104** to selected users.

The game system **212** provides various gaming functions within the context of the messaging client **104**. The messaging client **104** provides a game interface providing a list of available games that can be launched by a user within the context of the messaging client **104** and played with other users of the messaging system **100**. The messaging system **100** further enables a particular user to invite other users to participate in the play of a specific game, by issuing invitations to such other users from the messaging client **104**. The messaging client **104** also supports both voice and text messaging (e.g., chats) within the context of gameplay, provides a leaderboard for the games, and also supports the provision of in-game rewards (e.g., coins and items).

The external resource system **214** provides an interface for the messaging client **104** to communicate with remote servers (e.g., third-party servers **110**) to launch or access external resources, e.g., applications or applets. Each third-party server **110** hosts, for example, a markup language (e.g., HTML5) based application or small-scale version of an application (e.g., game, utility, payment, or ride-sharing application). The messaging client **104** may launch a web-based resource (e.g., application) by accessing the HTML5 file from the third-party servers **110** associated with the web-based resource. In certain examples, applications hosted by third-party servers **110** are programmed in JavaScript leveraging a Software Development Kit (SDK) provided by the messaging server **118**. The SDK includes Application Programming Interfaces (APIs) with functions that can be called or invoked by the web-based application. In certain examples, the messaging server **118** includes a JavaScript library that provides a given external resource access to certain user data of the messaging client **104**. HTML5 is used as an example technology for programming games, but applications and resources programmed based on other technologies can be used.

In order to integrate the functions of the SDK into the web-based resource, the SDK is downloaded by a third-party server **110** from the messaging server **118** or is otherwise received by the third-party server **110**. Once downloaded or received, the SDK is included as part of the application code of a web-based external resource. The code of the web-based resource can then call or invoke certain functions of the SDK to integrate features of the messaging client **104** into the web-based resource.

The SDK stored on the messaging server **118** effectively provides the bridge between an external resource (e.g., applications **106** or applets and the messaging client **104**). This provides the user with a seamless experience of communicating with other users on the messaging client **104**, while also preserving the look and feel of the messaging client **104**. To bridge communications between an external resource and a messaging client **104**, in certain examples, the SDK facilitates communication between third-party servers **110** and the messaging client **104**. In certain examples, a Web ViewJavaScriptBridge running on a client device **102** establishes two one-way communication channels between an external resource and the messaging client **104**. Messages are sent between the external resource and the messaging client **104** via these communication channels asynchronously. Each SDK function invocation is sent as a message and callback. Each SDK function is implemented by constructing a unique callback identifier and sending a message with that callback identifier.

By using the SDK, not all information from the messaging client **104** is shared with third-party servers **110**. The SDK limits which information is shared based on the needs of the external resource. In certain examples, each third-party server **110** provides an HTML5 file corresponding to the web-based external resource to the messaging server **118**. The messaging server **118** can add a visual representation (such as a box art or other graphic) of the web-based external resource in the messaging client **104**. Once the user selects the visual representation or instructs the messaging client **104** through a GUI of the messaging client **104** to access features of the web-based external resource, the messaging client **104** obtains the HTML5 file and instantiates the resources necessary to access the features of the web-based external resource.

The messaging client **104** presents a graphical user interface (e.g., a landing page or title screen) for an external

resource. During, before, or after presenting the landing section of reading materials such as a page or title screen, the messaging client **104** determines whether the launched external resource has been previously authorized to access user data of the messaging client **104**. In response to determining that the launched external resource has been previously authorized to access user data of the messaging client **104**, the messaging client **104** presents another graphical user interface of the external resource that includes functions and features of the external resource. In response to determining that the launched external resource has not been previously authorized to access user data of the messaging client **104**, after a threshold period of time (e.g., 3 seconds) of displaying the landing page or title screen of the external resource, the messaging client **104** slides up (e.g., animates a menu as surfacing from a bottom of the screen to a middle of or other portion of the screen) a menu for authorizing the external resource to access the user data. The menu identifies the type of user data that the external resource will be authorized to use. In response to receiving a user selection of an accept option, the messaging client **104** adds the external resource to a list of authorized external resources and allows the external resource to access user data from the messaging client **104**. In some examples, the external resource is authorized by the messaging client **104** to access the user data in accordance with an OAuth **2** framework.

The messaging client **104** controls the type of user data that is shared with external resources based on the type of external resource being authorized. For example, external resources that include full-scale applications (e.g., an application **106**) are provided with access to a first type of user data (e.g., only two-dimensional avatars of users with or without different avatar characteristics). As another example, external resources that include small-scale versions of applications (e.g., web-based versions of applications) are provided with access to a second type of user data (e.g., payment information, two-dimensional avatars of users, three-dimensional avatars of users, and avatars with various avatar characteristics). Avatar characteristics include different ways to customize a look and feel of an avatar, such as different poses, facial features, clothing, and so forth.

The tilt angle selection system **216** supports system **600** for AR wearable devices. The tilt angle selection system **216** may perform one or more of the functions described in conjunction with AR wearable device **602** herein. Additionally, the tilt angle selection system **216** may store the simulation data **622** and process the simulation data **622**. Additionally, the tilt angle selection system **216** may provide recommendations for a vertical tilt **618** angle for the AR display **616** based on which applications **604** and/or tasks **606** a user **608** is likely to use or has used.

The tilt angle selection system **216** receives requests from an AR wearable device **602** and responds to the requests. The requests include a request to process simulation data **622** and make recommendations regarding the vertical tilt **618** angle, which may include accompanying information as described in conjunction with FIG. **6**. The AR wearable devices **602** may request other services from the tilt angle selection system **216**.

Data Architecture

FIG. **3** is a schematic diagram illustrating data structures **300**, which may be stored in the database **126** of the messaging server system **108**, according to certain examples. While the content of the database **126** is shown to

comprise a number of tables, it will be appreciated that the data could be stored in other types of data structures (e.g., as an object-oriented database).

The database **126** includes message data stored within a message table **302**. This message data includes, for any particular one message, at least message sender data, message recipient (or receiver) data, and a payload. Further details regarding information that may be included in a message, and included within the message data stored in the message table **302** is described below with reference to FIG. **4**.

An entity table **306** stores entity data, and is linked (e.g., referentially) to an entity graph **308** and profile data **316**. Entities for which records are maintained within the entity table **306** may include individuals, corporate entities, organizations, objects, places, events, and so forth. Regardless of entity type, any entity regarding which the messaging server system **108** stores data may be a recognized entity. Each entity is provided with a unique identifier, as well as an entity type identifier (not shown).

The entity graph **308** stores information regarding relationships and associations between entities. Such relationships may be social, professional (e.g., work at a common corporation or organization) interested-based or activity-based, merely for example.

The profile data **316** stores multiple types of profile data about a particular entity. The profile data **316** may be selectively used and presented to other users of the messaging system **100**, based on privacy settings specified by a particular entity. Where the entity is an individual, the profile data **316** includes, for example, a user name, telephone number, address, settings (e.g., notification and privacy settings), as well as a user-selected avatar representation (or collection of such avatar representations). A particular user may then selectively include one or more of these avatar representations within the content of messages communicated via the messaging system **100**, and on map interfaces displayed by messaging clients **104** to other users. The collection of avatar representations may include "status avatars," which present a graphical representation of a status or activity that the user may select to communicate at a particular time.

Where the entity is a group, the profile data **316** for the group may similarly include one or more avatar representations associated with the group, in addition to the group name, members, and various settings (e.g., notifications) for the relevant group.

The database **126** also stores augmentation data, such as overlays or filters, in an augmentation table **310**. The augmentation data is associated with and applied to videos (for which data is stored in a video table **304**) and images (for which data is stored in an image table **312**).

Filters, in one example, are overlays that are displayed as overlaid on an image or video during presentation to a recipient user. Filters may be of various types, including user-selected filters from a set of filters presented to a sending user by the messaging client **104** when the sending user is composing a message. Other types of filters include geolocation filters (also known as geo-filters), which may be presented to a sending user based on geographic location. For example, geolocation filters specific to a neighborhood or special location may be presented within a user interface by the messaging client **104**, based on geolocation information determined by a Global Positioning System (GPS) unit of the client device **102**.

Another type of filter is a data filter, which may be selectively presented to a sending user by the messaging

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client 104, based on other inputs or information gathered by the client device 102 during the message creation process. Examples of data filters include current temperature at a specific location, a current speed at which a sending user is traveling, battery life for a client device 102, or the current time.

Other augmentation data that may be stored within the image table 312 includes augmented reality content items (e.g., corresponding to applying Lenses or augmented reality experiences). An augmented reality content item may be a real-time special effect and sound that may be added to an image or a video.

As described above, augmentation data includes augmented reality content items, overlays, image transformations, AR images, and similar terms refer to modifications that may be applied to image data (e.g., videos or images). This includes real-time modifications, which modify an image as it is captured using device sensors (e.g., one or multiple cameras) of a client device 102 and then displayed on a screen of the client device 102 with the modifications. This also includes modifications to stored content, such as video clips in a gallery that may be modified. For example, in a client device 102 with access to multiple augmented reality content items, a user can use a single video clip with multiple augmented reality content items to see how the different augmented reality content items will modify the stored clip. For example, multiple augmented reality content items that apply different pseudorandom movement models can be applied to the same content by selecting different augmented reality content items for the content. Similarly, real-time video capture may be used with an illustrated modification to show how video images currently being captured by sensors of a client device 102 would modify the captured data. Such data may simply be displayed on the screen and not stored in memory, or the content captured by the device sensors may be recorded and stored in memory with or without the modifications (or both). In some systems, a preview feature can show how different augmented reality content items will look within different windows in a display at the same time. This can, for example, enable multiple windows with different pseudorandom animations to be viewed on a display at the same time.

Data and various systems using augmented reality content items or other such transform systems to modify content using this data can thus involve detection of objects (e.g., faces, hands, bodies, cats, dogs, surfaces, objects, etc.), tracking of such objects as they leave, enter, and move around the field of view in video frames, and the modification or transformation of such objects as they are tracked. In various examples, different methods for achieving such transformations may be used. Some examples may involve generating a three-dimensional mesh model of the object or objects and using transformations and animated textures of the model within the video to achieve the transformation. In other examples, tracking of points on an object may be used to place an image or texture (which may be two dimensional or three dimensional) at the tracked position. In still further examples, neural network analysis of video frames may be used to place images, models, or textures in content (e.g., images or frames of video). Augmented reality content items thus refer both to the images, models, and textures used to create transformations in content, as well as to additional modeling and analysis information needed to achieve such transformations with object detection, tracking, and placement.

Real-time video processing can be performed with any kind of video data (e.g., video streams, video files, etc.)

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saved in a memory of a computerized system of any kind. For example, a user can load video files and save them in a memory of a device, or can generate a video stream using sensors of the device. Additionally, any objects can be processed using a computer animation model, such as a human's face and parts of a human body, animals, or non-living things such as chairs, cars, or other objects.

In some examples, when a particular modification is selected along with content to be transformed, elements to be transformed are identified by the computing device, and then detected and tracked if they are present in the frames of the video. The elements of the object are modified according to the request for modification, thus transforming the frames of the video stream. Transformation of frames of a video stream can be performed by different methods for different kinds of transformation. For example, for transformations of frames mostly referring to changing forms of object's elements characteristic points for each element of an object are calculated (e.g., using an Active Shape Model (ASM) or other known methods). Then, a mesh based on the characteristic points is generated for each of the at least one element of the object. This mesh is used in the following stage of tracking the elements of the object in the video stream. In the process of tracking, the mentioned mesh for each element is aligned with a position of each element. Then, additional points are generated on the mesh. A first set of first points is generated for each element based on a request for modification, and a set of second points is generated for each element based on the set of first points and the request for modification. Then, the frames of the video stream can be transformed by modifying the elements of the object on the basis of the sets of first and second points and the mesh. In such method, a background of the modified object can be changed or distorted as well by tracking and modifying the background.

In some examples, transformations changing some areas of an object using its elements can be performed by calculating characteristic points for each element of an object and generating a mesh based on the calculated characteristic points. Points are generated on the mesh, and then various areas based on the points are generated. The elements of the object are then tracked by aligning the area for each element with a position for each of the at least one element, and properties of the areas can be modified based on the request for modification, thus transforming the frames of the video stream. Depending on the specific request for modification properties of the mentioned areas can be transformed in different ways. Such modifications may involve changing color of areas; removing at least some part of areas from the frames of the video stream; including one or more new objects into areas which are based on a request for modification; and modifying or distorting the elements of an area or object. In various examples, any combination of such modifications or other similar modifications may be used. For certain models to be animated, some characteristic points can be selected as control points to be used in determining the entire state-space of options for the model animation.

In some examples of a computer animation model to transform image data using face detection, the face is detected on an image with use of a specific face detection algorithm (e.g., Viola-Jones). Then, an Active Shape Model (ASM) algorithm is applied to the face region of an image to detect facial feature reference points.

Other methods and algorithms suitable for face detection can be used. For example, in some examples, features are located using a landmark, which represents a distinguishable

point present in most of the images under consideration. For facial landmarks, for example, the location of the left eye pupil may be used. If an initial landmark is not identifiable (e.g., if a person has an eyepatch), secondary landmarks may be used. Such landmark identification procedures may be used for any such objects. In some examples, a set of landmarks forms a shape. Shapes can be represented as vectors using the coordinates of the points in the shape. One shape is aligned to another with a similarity transform (allowing translation, scaling, and rotation) that minimizes the average Euclidean distance between shape points. The mean shape is the mean of the aligned training shapes.

In some examples, a search for landmarks from the mean shape aligned to the position and size of the face determined by a global face detector is started. Such a search then repeats the steps of suggesting a tentative shape by adjusting the locations of shape points by template matching of the image texture around each point and then conforming the tentative shape to a global shape model until convergence occurs. In some systems, individual template matches are unreliable, and the shape model pools the results of the weak template matches to form a stronger overall classifier. The entire search is repeated at each level in an image pyramid, from coarse to fine resolution.

A transformation system can capture an image or video stream on a client device (e.g., the client device 102) and perform complex image manipulations locally on the client device 102 while maintaining a suitable user experience, computation time, and power consumption. The complex image manipulations may include size and shape changes, emotion transfers (e.g., changing a face from a frown to a smile), state transfers (e.g., aging a subject, reducing apparent age, changing gender), style transfers, graphical element application, and any other suitable image or video manipulation implemented by a convolutional neural network that has been configured to execute efficiently on the client device 102.

In some examples, a computer animation model to transform image data can be used by a system where a user may capture an image or video stream of the user (e.g., a selfie) using a client device 102 having a neural network operating as part of a messaging client 104 operating on the client device 102. The transformation system operating within the messaging client 104 determines the presence of a face within the image or video stream and provides modification icons associated with a computer animation model to transform image data, or the computer animation model can be present as associated with an interface described herein. The modification icons include changes that may be the basis for modifying the user's face within the image or video stream as part of the modification operation. Once a modification icon is selected, the transform system initiates a process to convert the image of the user to reflect the selected modification icon (e.g., generate a smiling face on the user). A modified image or video stream may be presented in a graphical user interface displayed on the client device 102 as soon as the image or video stream is captured, and a specified modification is selected. The transformation system may implement a complex convolutional neural network on a portion of the image or video stream to generate and apply the selected modification. That is, the user may capture the image or video stream and be presented with a modified result in real-time or near real-time once a modification icon has been selected. Further, the modification may be persistent while the video stream is being captured,

and the selected modification icon remains toggled. Machine taught neural networks may be used to enable such modifications.

The graphical user interface, presenting the modification performed by the transform system, may supply the user with additional interaction options. Such options may be based on the interface used to initiate the content capture and selection of a particular computer animation model (e.g., initiation from a content creator user interface). In various examples, a modification may be persistent after an initial selection of a modification icon. The user may toggle the modification on or off by tapping or otherwise selecting the face being modified by the transformation system and store it for later viewing or browse to other areas of the imaging application. Where multiple faces are modified by the transformation system, the user may toggle the modification on or off globally by tapping or selecting a single face modified and displayed within a graphical user interface. In some examples, individual faces, among a group of multiple faces, may be individually modified, or such modifications may be individually toggled by tapping or selecting the individual face or a series of individual faces displayed within the graphical user interface.

A story table 314 stores data regarding collections of messages and associated image, video, or audio data, which are compiled into a collection (e.g., a story or a gallery). The creation of a particular collection may be initiated by a particular user (e.g., each user for which a record is maintained in the entity table 306). A user may create a "personal story" in the form of a collection of content that has been created and sent/broadcast by that user. To this end, the user interface of the messaging client 104 may include an icon that is user-selectable to enable a sending user to add specific content to his or her personal story.

A collection may also constitute a "live story," which is a collection of content from multiple users that is created manually, automatically, or using a combination of manual and automatic techniques. For example, a "live story" may constitute a curated stream of user-submitted content from varies locations and events. Users whose client devices have location services enabled and are at a common location event at a particular time may, for example, be presented with an option, via a user interface of the messaging client 104, to contribute content to a particular live story. The live story may be identified to the user by the messaging client 104, based on his or her location. The end result is a "live story" told from a community perspective.

A further type of content collection is known as a "location story," which enables a user whose client device 102 is located within a specific geographic location (e.g., on a college or university campus) to contribute to a particular collection. In some examples, a contribution to a location story may require a second degree of authentication to verify that the end-user belongs to a specific organization or other entity (e.g., is a student on the university campus).

As mentioned above, the video table 304 stores video data that, in one example, is associated with messages for which records are maintained within the message table 302. Similarly, the image table 312 stores image data associated with messages for which message data is stored in the entity table 306. The entity table 306 may associate various augmentations from the augmentation table 310 with various images and videos stored in the image table 312 and the video table 304.

Data Communications Architecture

FIG. 4 is a schematic diagram illustrating a structure of a message 400, according to some examples, generated by a

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messaging client **104** for communication to a further messaging client **104** or the messaging server **118**. The content of a particular message **400** is used to populate the message table **302** stored within the database **126**, accessible by the messaging server **118**. Similarly, the content of a message **400** is stored in memory as “in-transit” or “in-flight” data of the client device **102** or the application servers **114**. A message **400** is shown to include the following example components:

message identifier **402**: a unique identifier that identifies the message **400**.

message text payload **404**: text, to be generated by a user via a user interface of the client device **102**, and that is included in the message **400**.

message image payload **406**: image data, captured by a camera component of a client device **102** or retrieved from a memory component of a client device **102**, and that is included in the message **400**. Image data for a sent or received message **400** may be stored in the image table **312**.

message video payload **408**: video data, captured by a camera component or retrieved from a memory component of the client device **102**, and that is included in the message **400**. Video data for a sent or received message **400** may be stored in the video table **304**.

message audio payload **410**: audio data, captured by a microphone or retrieved from a memory component of the client device **102**, and that is included in the message **400**.

message augmentation data **412**: augmentation data (e.g., filters, stickers, or other annotations or enhancements) that represents augmentations to be applied to message image payload **406**, message video payload **408**, or message audio payload **410** of the message **400**. Augmentation data for a sent or received message **400** may be stored in the augmentation table **310**.

message duration parameter **414**: parameter value indicating, in seconds, the amount of time for which content of the message (e.g., the message image payload **406**, message video payload **408**, message audio payload **410**) is to be presented or made accessible to a user via the messaging client **104**.

message geolocation parameter **416**: geolocation data (e.g., latitudinal and longitudinal coordinates) associated with the content payload of the message. Multiple message geolocation parameter **416** values may be included in the payload, each of these parameter values being associated with respect to content items included in the content (e.g., a specific image into within the message image payload **406**, or a specific video in the message video payload **408**).

message story identifier **418**: identifier values identifying one or more content collections (e.g., “stories” identified in the story table **314**) with which a particular content item in the message image payload **406** of the message **400** is associated. For example, multiple images within the message image payload **406** may each be associated with multiple content collections using identifier values.

message tag **420**: each message **400** may be tagged with multiple tags, each of which is indicative of the subject matter of content included in the message payload. For example, where a particular image included in the message image payload **406** depicts an animal (e.g., a lion), a tag value may be included within the message tag **420** that is indicative of the relevant animal. Tag

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values may be generated manually, based on user input, or may be automatically generated using, for example, image recognition.

message sender identifier **422**: an identifier (e.g., a messaging system identifier, email address, or device identifier) indicative of a user of the Client device **102** on which the message **400** was generated and from which the message **400** was sent.

message receiver identifier **424**: an identifier (e.g., a messaging system identifier, email address, or device identifier) indicative of a user of the client device **102** to which the message **400** is addressed.

The contents (e.g., values) of the various components of message **400** may be pointers to locations in tables within which content data values are stored. For example, an image value in the message image payload **406** may be a pointer to (or address of) a location within an image table **312**. Similarly, values within the message video payload **408** may point to data stored within a video table **304**, values stored within the message augmentations **412** may point to data stored in an augmentation table **310**, values stored within the message story identifier **418** may point to data stored in a story table **314**, and values stored within the message sender identifier **422** and the message receiver identifier **424** may point to user records stored within an entity table **306**.

Time-Based Access Limitation Architecture

FIG. **5** is a schematic diagram illustrating an access-limiting process **500**, in terms of which access to content (e.g., an ephemeral message **502**, and associated multimedia payload of data) or a content collection (e.g., an ephemeral message group **504**) may be time-limited (e.g., made ephemeral).

An ephemeral message **502** is shown to be associated with a message duration parameter **506**, the value of which determines an amount of time that the ephemeral message **502** will be displayed to a receiving user of the ephemeral message **502** by the messaging client **104**. In one example, an ephemeral message **502** is viewable by a receiving user for up to a maximum of 10 seconds, depending on the amount of time that the sending user specifies using the message duration parameter **506**.

The message duration parameter **506** and the message receiver identifier **424** are shown to be inputs to a message timer **510**, which is responsible for determining the amount of time that the ephemeral message **502** is shown to a particular receiving user identified by the message receiver identifier **424**. In particular, the ephemeral message **502** will only be shown to the relevant receiving user for a time period determined by the value of the message duration parameter **506**. The message timer **510** is shown to provide output to a more generalized ephemeral timer system **202**, which is responsible for the overall timing of display of content (e.g., an ephemeral message **502**) to a receiving user.

The ephemeral message **502** is shown in FIG. **5** to be included within an ephemeral message group **504** (e.g., a collection of messages in a personal story, or an event story). The ephemeral message group **504** has an associated group duration parameter **508**, a value of which determines a time duration for which the ephemeral message group **504** is presented and accessible to users of the messaging system **100**. The group duration parameter **508**, for example, may be the duration of a music concert, where the ephemeral message group **504** is a collection of content pertaining to that concert. Alternatively, a user (either the owning user or a curator user) may specify the value for the group duration parameter **508** when performing the setup and creation of the ephemeral message group **504**.

Additionally, each ephemeral message **502** within the ephemeral message group **504** has an associated group participation parameter **512**, a value of which determines the duration of time for which the ephemeral message **502** will be accessible within the context of the ephemeral message group **504**. Accordingly, a particular ephemeral message group **504** may “expire” and become inaccessible within the context of the ephemeral message group **504**, prior to the ephemeral message group **504** itself expiring in terms of the group duration parameter **508**. The group duration parameter **508**, group participation parameter **512**, and message receiver identifier **424** each provide input to a group timer **514**, which operationally determines, firstly, whether a particular ephemeral message **502** of the ephemeral message group **504** will be displayed to a particular receiving user and, if so, for how long. Note that the ephemeral message group **504** is also aware of the identity of the particular receiving user as a result of the message receiver identifier **424**.

Accordingly, the group timer **514** operationally controls the overall lifespan of an associated ephemeral message group **504**, as well as an individual ephemeral message **502** included in the ephemeral message group **504**. In one example, each and every ephemeral message **502** within the ephemeral message group **504** remains viewable and accessible for a time period specified by the group duration parameter **508**. In a further example, a certain ephemeral message **502** may expire, within the context of ephemeral message group **504**, based on a group participation parameter **512**. Note that a message duration parameter **506** may still determine the duration of time for which a particular ephemeral message **502** is displayed to a receiving user, even within the context of the ephemeral message group **504**. Accordingly, the message duration parameter **506** determines the duration of time that a particular ephemeral message **502** is displayed to a receiving user, regardless of whether the receiving user is viewing that ephemeral message **502** inside or outside the context of an ephemeral message group **504**.

The ephemeral timer system **202** may furthermore operationally remove a particular ephemeral message **502** from the ephemeral message group **504** based on a determination that it has exceeded an associated group participation parameter **512**. For example, when a sending user has established a group participation parameter **512** of 24 hours from posting, the ephemeral timer system **202** will remove the relevant ephemeral message **502** from the ephemeral message group **504** after the specified 24 hours. The ephemeral timer system **202** also operates to remove an ephemeral message group **504** when either the group participation parameter **512** for each and every ephemeral message **502** within the ephemeral message group **504** has expired, or when the ephemeral message group **504** itself has expired in terms of the group duration parameter **508**.

In certain use cases, a creator of a particular ephemeral message group **504** may specify an indefinite group duration parameter **508**. In this case, the expiration of the group participation parameter **512** for the last remaining ephemeral message **502** within the ephemeral message group **504** will determine when the ephemeral message group **504** itself expires. In this case, a new ephemeral message **502**, added to the ephemeral message group **504**, with a new group participation parameter **512**, effectively extends the life of an ephemeral message group **504** to equal the value of the group participation parameter **512**.

Responsive to the ephemeral timer system **202** determining that an ephemeral message group **504** has expired (e.g.,

is no longer accessible), the ephemeral timer system **202** communicates with the messaging system **100** (and, for example, specifically the messaging client **104**) to cause an indicium (e.g., an icon) associated with the relevant ephemeral message group **504** to no longer be displayed within a user interface of the messaging client **104**. Similarly, when the ephemeral timer system **202** determines that the message duration parameter **506** for a particular ephemeral message **502** has expired, the ephemeral timer system **202** causes the messaging client **104** to no longer display an indicium (e.g., an icon or textual identification) associated with the ephemeral message **502**.

Selecting a Tilt Angle of an AR Display

FIG. 6 illustrates a system **600** for selecting a tilt angle of an AR display, in accordance with some examples. The system **600** includes an AR wearable device **602** such as glasses **1200** of FIG. 12 and may include other devices such as a portion of the messaging server system **108** or the client device **102** that may perform one or more of the operations described herein. The term AR graphics includes anything displayed by the AR display **616** for the user **608** to view in conjunction with viewing the real-world through lenses. The AR display **616** cannot display graphics **646** over the full gazing range **645**, which is an up and down range, of a user **608**. The system **600** determines a vertical tilt **618** or tilt angle to use for the AR display **616**, which determines where the AR graphics view **644** falls within the gazing range **645** of the user **608**. The vertical tilt **618** may be between +90 degrees and -90 degrees. In some examples, the vertical tilt **618** ranges from +30 degrees to -40 degrees. As an example, the gazing range **645** may be 70 degrees, which indicates the range which the user **608** moves their eyes up and down to look at something without moving their head and thus the AR wearable device **602**. The AR graphics view **644** may be only 27.4 degrees or less than half the full gazing range **645** of the user **608**. The AR wearable device **602** thus cannot display graphics **646** on the AR display **616** over the full gazing range **645** of the user **608** and the AR graphics view **644** is determined by the selected vertical tilt **618** selected for the AR display **616**. The AR wearable device **602** is worn by the user **608**.

The display **610** includes optical elements **612** and the AR display **616**. The user **608** sees a user view **642** of the real-world scene **634** through the optical elements **612** and an AR graphics view **644** from the graphics **646** displayed by the AR display **616**, in accordance with some examples. The AR display **616** includes a projector that projects the graphics **646** of the AR graphics view **644** on the optical elements **612**, in accordance with some examples. One skilled in the art would recognize that other AR display **616** designs may be used and that other designs may be used that do not include optical elements **612**. The AR display **616** includes a field of view **636** and a vertical tilt **618**. The field of view **636** may be characterized by a degree such as 27.4 degrees, which indicates the range in which the AR display **616** can display graphics **646** on the display **610**. The field of view **636** is symmetric about the vertical tilt **618**. The field of view **636** is narrower than the fixed field of view **640** of the optical elements **612**. The user view **642** corresponds to what the user **608** can see of the real-world scene **634** through the optical elements **612**. The AR graphics view **644** corresponds to the graphics **646** seen by the user **608** where the graphics **646** are displayed by the AR display **616**.

The optical elements **612** have a vertical tilt **614**, which may be zero degrees, but may be set at a downward angle of 1 degree to 20 degrees or another number of degrees. The AR display **616** has a vertical tilt **618**, which may be zero

degrees, but may be set at a downward angle of 1 degree to 40 degrees or another number of degrees. The system 600 determines a vertical tilt 618 to use for the AR display 616.

Applications 604 are run on the AR wearable device 602. Example applications 604 include applications 604 that use AR graphics to assist users in tasks they are performing, applications 604 that identify real-world objects, applications 604 that provide directions, and so forth. A user 608 performs tasks 606 with the applications 604. For example, a user 608 may use an application 604 that assists the user 608 in selecting a breakfast cereal in the grocery store. The task 606 is to select the breakfast cereal. The graphics 646 may indicate a rating of the breakfast cereal in terms of nutrition and sugar content, past purchases of the user 608, and any rating of the breakfast cereal by the user 608 or members of a family of the user 608. The viewing gaze range 638 indicates the viewing gaze range 638 of a user 608 in performing the task 606. For example, the viewing gaze range 638 may be 60 degrees for the task 606 of selecting a breakfast cereal as the user 608 looks up and down different shelves in the aisle of a grocery store. The viewing gaze range 638 may be larger than the field of view 636. For example, the user 608 may look up at the top shelf and down at a bottom shelf without moving their heads so that the viewing gaze range 638 may be large such as 60 degrees. But the field of view 636 may be only 27.4 degrees so that the AR display 616 cannot display graphics 646 in all the places the user 608 may look or gaze. The viewing gaze range 638 may be determined by actually measuring the viewing gaze range 638 of a number of users 608 in performing the task 606.

The simulator module 620 first accesses simulation data 622. The simulation data 622 is gathered by performing studies or experiments where users 608 perform tasks 606 of applications 604 and where the viewing gaze range 638 used in performing the tasks 606 is measured. The simulation data 622 may be gathered in usability laboratories where the viewing gaze range 638 is monitored or tracked with equipment separate from the AR wearable device 602. Alternatively, or in addition, the simulation data 622 may be gathered by an instrument attached to or part of the AR wearable device 602 that monitors or tracks the viewing gaze of the user 608. In some examples, the simulator module 620 is configured to monitor the viewing gaze of the user 608 and collect the simulation data 622. In the case of the AR wearable device 602 being able to track the viewing gaze angle 626, then the simulation data 622 can be gathered from users 608 that are using the AR wearable device 602 and stored in at a central server such as at the messaging server system 108.

The simulation data 622 indicates the application 624, the task 606, the vertical gaze angle 626 for a gaze fixation of the user 628, and the user 628. For example, the application 624 is the application 604 and task 606 are selecting a morning cereal. The vertical gaze angle 626 is an angle (up or down) of a gaze fixation of the user 628 while the user 628 is performing the task 606 of selecting a breakfast cereal. The vertical gaze angle 626 indicates a gaze fixation was detected at the vertical gaze angle 626 such as -10 degrees or 20 degrees. A gaze fixation is when the eye 802 of the user 608 fixates on a location for at least a threshold hold period of time such as 0.01 seconds or another number.

The simulator module 620 processes the simulation data 622 to generate the processed simulation data 630. For example, the processed simulation data 630 includes a probability distribution of vertical viewing angles for the selected applications 624 and tasks 606. Graph 1000 pro-

vides an example probability distribution of vertical viewing angles. The simulator module 620 may process the simulation data 622 to generate other statistical or probabilistic information. For example, there may be a task 606 of an application 604 that is considered to be important to support. The simulator module 620 may weigh the viewing gaze range 638 of the task 606 that is important to support more heavily in creating the processed simulation data 630. The view gaze range 638 may be indicated by a distribution of a number of gaze fixations per vertical gaze angles.

The select AR display tilt module 632 selects the vertical tilt 618 to use. In some examples, the select AR display tilt module 632 sweeps through all possible vertical tilt 618 angles and determines what fraction of the total distribution, of the probability distribution of vertical viewing angles, would fall within the field of view 636 for each vertical tilt 618 angle. The select AR display tilt module 632 then selects the vertical tilt 618 angle that maximizes the fraction of gaze fixations that are included in the field of view 636. The select AR display tilt module 632 may select the vertical tilt 618 to ensure a particular vertical gaze angle 626 is included in the field of view 636 such as -10 degrees in order to support a task 606 of an important application 604. In some examples, the vertical tilt 618 angle is selected and then the AR wearable device 602 is manufactured with a permanent vertical tilt 618 relative to the vertical tilt 614 of the optical elements 612. In some examples, sources of a noise distribution are accounted for. An example of noise is users 608 wearing the AR wearable device 602 differently or wearing the AR wearable device 602 differently at different times, which may change the vertical tilt 614 of the optical elements 612. The simulator module 620 may convolve a distribution of a source of noise with the simulation data 622 when generating the processed simulation data 630. The processed simulation data 630 then includes the source of noise and will affect the selection of the vertical tilt 618 of the AR display 616. For example, if a user 608 wears the AR wearable device 602 correctly or as intended 80 percent of the time but at an extra vertical tilt 614 of 10 degrees 20 percent of the time, then the simulator module 620 shifts the raw data with 20 percent of the raw data shifted by 10 degrees for the vertical gaze angle 626.

FIG. 7 illustrates a system 700 for selecting a tilt angle of an AR display, in accordance with some examples. The system 700 includes an AR wearable device 602 such as glasses 1200 of FIG. 12 and may include other devices such as a portion of the messaging server system 108 or the client device 102 that may perform one or more of the operations described herein.

In some examples, the vertical tilt 618 angle is adjustable. The select AR display tilt module 632 resides on the AR wearable device 602, the messaging server system 108, the client device 102, or another device. The select AR display tilt module 632 selects the vertical tilt 618 angle based on the applications 604 and tasks 606 the user 608 is using or likely will use and causes an actuator (not illustrated) to adjust the vertical tilt 618 angle.

FIG. 8 illustrates the vertical tilt 618 angle, in accordance with some examples. The eye 802 is of a user 608 wearing the AR wearable device 602. The optical element 612 is tilted at a vertical tilt 614 angle relative to the horizon 804 or a normal line coming out of the eye 802. The vertical tilt 618 angle indicates the angle between the horizon 804 and the AR display 616 (not illustrated in FIG. 8). The AR display 616 may be above mounted on the AR wearable device 602 or mounted on or a structure associated with the optical element 612. One skilled in the art would recognize

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that the AR display 616 may be mounted in different ways. Additionally, in some examples, the AR display 616 may be realized differently such as via elements within the optical element 612 or directly projecting onto the eye 802. The selection of the field of view 636 is the same or similar with different AR displays 616 as the selection of the vertical tilt 618 angle.

FIG. 9 illustrates the vertical tilt 618 angle, in accordance with some examples. The person 902 is wearing the AR wearable device 602. The field of view 636 is determined by the vertical tilt 618 angle in relation to the horizon 804. The AR wearable device 602 cannot project graphics 646 on object 1 904 because it is outside the field of view 636. So, if the eye 802 of the person 902 looks up or down without moving the head of the person 902, then object 1 904 or object 3 908, respectively, will appear as it does in the real-world scene 634 without any graphics 646. However, since object 2 906 is within the field of view 636 of the AR display 616, the AR wearable device 602 can generate graphics 910 associated with object 2 906. For example, the graphics 910 may be projected onto object 2 906, which may be a cereal box. The graphics 910 may be adjusted in size and perspective to appear to complement the face of the cereal box to add information such as a likelihood the person 902 will like the cereal and a health score for cereal.

FIG. 10 illustrates a graph 1000, in accordance with some examples. The graph 1000 graphs a number of gaze fixations 1008 for vertical gaze angles 1010. In some examples, vertical gaze angles range from +30 degrees to -40 degrees. This graph 1000 is similar to a graph of a probability distribution of vertical viewing angles discussed in conjunction with FIG. 6 except a raw number of gaze fixations 1008 are used rather than a probability of a gaze fixation. The gaze fixation 1008 is a number of gaze fixations times 10^4 that were counted when users 608 were performing tasks 606 of applications 604 that are covered by the graph 1000. The vertical tilt 618 illustrated in the graph 1000 is -4.6 degrees with respect to the horizon 804. The gaze in field of view (FOV) 1004 is AR graphics view 644. The gaze out of FOV 1002 is user view 642 which is not part of the gaze in FOV 1004. For example, the object 1 904 would in the above 1014 portion of the graph 1000 and the object 3 908 would be in the below portion 1012 of the graph. The object 2 906 would be within the field of view 636 of the AR display 616. The select AR display tilt module 632 may select the bore sight or vertical tilt 618 to maximize the area under the curve. The select AR display tilt module 632 may select the bore sight or vertical tilt 618 with the largest number of gaze fixations 1008 within the gaze in FOV 1004. In some examples, the vertical tilt 618 is adjusted to accommodate important or necessary gaze fixations 1008. For example, a gaze fixation of -22 degree may be the gaze used to check the battery level of the AR wearable device 602 so the select AR display tilt module 632 ensures that -22 degrees is within the gaze in FOV 1004.

FIG. 11 illustrates a graph 1100, in accordance with some examples. The graph 1100 graphs the fraction in the FOV 1102, which refers to the fraction of gaze fixations 1008 of FIG. 10, against the boresight angle 1104 or the vertical tilt 618. The graph 1100 is effectively the gaze in FOV 1004 of FIG. 10 divided by the total number of gaze fixations of FIG. 10 based on the vertical tilt 618. The maximum inclusion 1108 is the boresight angle 1104 where the fraction in FOV 1102 is highest, which here is about 9 degrees.

FIG. 12 is a perspective view of a wearable electronic device in the form of glasses 1200, in accordance with some examples. The glasses 1200 are an article of eyewear

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including electronics, which operate within a network system for communicating image and video content. In some examples, the wearable electronic device is termed AR glasses. The glasses 1200 can include a frame 1232 made from any suitable material such as plastic or metal, including any suitable shape memory alloy. The frame 1232 can have a front piece 1233 that can include a first or left lens, display, or optical element holder 1236 and a second or right lens, display, or optical element holder 1237 connected by a bridge 1238. The front piece 1233 additionally includes a left end portion 1241 and a right end portion 1242. A first or left optical element 1244 and a second or right optical element 1243 can be provided within respective left and right optical element holders 1236, 1237. Each of the optical elements 1243, 1244 can be a lens, a display, a display assembly, or a combination of the foregoing. In some examples, for example, the glasses 1200 are provided with an integrated near-eye display mechanism that enables, for example, display to the user of preview images for visual media captured by cameras 1269 of the glasses 1200.

The frame 1232 additionally includes a left arm or temple piece 1246 and a right arm or temple piece 1247 coupled to the respective left and right end portions 1241, 1242 of the front piece 1233 by any suitable means such as a hinge (not shown), so as to be coupled to the front piece 1233, or rigidly or fixedly secured to the front piece 1233 so as to be integral with the front piece 1233. Each of the temple pieces 1246 and 1247 can include a first portion 1251 that is coupled to the respective end portion 1241 or 1242 of the front piece 1233 and any suitable second portion 1252, such as a curved or arcuate piece, for coupling to the ear of the user. In one example, the front piece 1233 can be formed from a single piece of material, so as to have a unitary or integral construction. In one example, the entire frame 1232 can be formed from a single piece of material so as to have a unitary or integral construction.

The glasses 1200 include a computing device, such as a computer 1261, which can be of any suitable type so as to be carried by the frame 1232 and, in one example, of a suitable size and shape, so as to be at least partially disposed in one or more of the temple pieces 1246 and 1247. In one example, the computer 1261 has a size and shape similar to the size and shape of one of the temple pieces 1246, 1247 and is thus disposed almost entirely if not entirely within the structure and confines of such temple pieces 1246 and 1247.

In one example, the computer 1261 can be disposed in both of the temple pieces 1246, 1247. The computer 1261 can include one or more processors with memory, wireless communication circuitry, and a power source. The computer 1261 comprises low-power circuitry, high-speed circuitry, location circuitry, and a display processor. Various other examples may include these elements in different configurations or integrated together in different ways. Additional details of aspects of the computer 1261 may be implemented as described with reference to the description that follows.

The computer 1261 additionally includes a battery 1262 or other suitable portable power supply. In one example, the battery 1262 is disposed in one of the temple pieces 1246 or 1247. In the glasses 1200 shown in FIG. 12, the battery 1262 is shown as being disposed in the left temple piece 1246 and electrically coupled using a connection 1274 to the remainder of the computer 1261 disposed in the right temple piece 1247. One or more input and output devices can include a connector or port (not shown) suitable for charging a battery 1262 accessible from the outside of the frame 1232, a wireless receiver, transmitter, or transceiver (not shown), or a combination of such devices.

The glasses **1200** include digital cameras **1269**. Although two cameras **1269** are depicted, other examples contemplate the use of a single or additional (i.e., more than two) cameras **1269**. For ease of description, various features relating to the cameras **1269** will be described further with reference to only a single camera **1269**, but it will be appreciated that these features can apply, in suitable examples, to both cameras **1269**.

In various examples, the glasses **1200** may include any number of input sensors or peripheral devices in addition to the cameras **1269**. The front piece **1233** is provided with an outward-facing, forward-facing, front, or outer surface **1266** that faces forward or away from the user when the glasses **1200** are mounted on the face of the user, and an opposite inward-facing, rearward-facing, rear, or inner surface **1267** that faces the face of the user when the glasses **1200** are mounted on the face of the user. Such sensors can include inward-facing video sensors or digital imaging modules such as cameras **1269** that can be mounted on or provided within the inner surface **1267** of the front piece **1233** or elsewhere on the frame **1232** so as to be facing the user, and outward-facing video sensors or digital imaging modules such as the cameras **1269** that can be mounted on or provided with the outer surface **1266** of the front piece **1233** or elsewhere on the frame **1232** so as to be facing away from the user. Such sensors, peripheral devices, or peripherals can additionally include biometric sensors, location sensors, accelerometers, or any other such sensors. In some examples, projectors (not illustrated) are used to project images on the inner surface of the optical elements **1243**, **1244** (or lenses) to provide a mixed reality or augmented reality experience for the user of the glasses **1200**.

The glasses **1200** further include an example of a camera control mechanism or user input mechanism comprising a camera control button mounted on the frame **1232** for haptic or manual engagement by the user. The camera control button provides a bi-modal or single-action mechanism in that it is disposable by the user between only two conditions, namely an engaged condition and a disengaged condition. In this example, the camera control button is a push button that is by default in the disengaged condition, being depressible by the user to dispose it to the engaged condition. Upon release of the depressed camera control button, it automatically returns to the disengaged condition.

In other examples, the single-action input mechanism can instead be provided by, for example, a touch-sensitive button comprising a capacitive sensor mounted on the frame **1232** adjacent to its surface for detecting the presence of a user's finger, to dispose the touch-sensitive button to the engaged condition when the user touches a finger to the corresponding spot on the outer surface **1266** of the frame **1232**. It will be appreciated that the above-described camera control button and capacitive touch button are but two examples of a haptic input mechanism for single-action control of the camera **1269**, and that other examples may employ different single-action haptic control arrangements.

The computer **1261** is configured to perform the methods described herein. In some examples, the computer **1261** is coupled to one or more antennas for reception of signals from a GNSS and circuitry for processing the signals where the antennas and circuitry are housed in the glasses **1200**. In some examples, the computer **1261** is coupled to one or more wireless antennas and circuitry for transmitting and receiving wireless signals where the antennas and circuitry are housed in the glasses **1200**. In some examples, there are multiple sets of antennas and circuitry housed in the glasses **1200**. In some examples, the antennas and circuitry are

configured to operate in accordance with a communication protocol such as Bluetooth™, Low-energy Bluetooth™, IEEE 802, IEEE 802.11az/be, and so forth. In some examples, PDR sensors housed in glasses **1200** and coupled to the computer **1261**. In some examples, the glasses **1200** are VR headsets where optical elements **1243**, **1244** are opaque screens for displaying images to a user of the VR headset. In some examples, the computer **1261** is coupled to user interface elements such as slide or touchpad **1276** and button **1278**. A long press of button **1278** resets the glasses **1200**. The slide or touchpad **1276** and button **1278** are used for a user to provide input to the computer **1261** and/or other electronic components of the glasses **1200**. The glasses **1200** include one or more microphones **1282** that are coupled to the computer **1261**. The glasses **1200** include one or more gyroscopes **1280**. The glasses **1200** include an AR display **616** as described herein.

FIG. **13** illustrates a method **1300** for selecting a tilt angle of an AR display, in accordance with some examples. The method **1300** may be performed by an AR wearable device **602** such as glasses **1200** of FIG. **12**. The method **1300** may be performed in other ways to generate the tilt angle for manufacturing the AR display **616** and/or the AR wearable device **602**. One or more of the operations may be performed wholly or in part by other devices such as a portion of the messaging server system **108**, the client device **102**, or in another way.

The method **1300** for setting a tilt angle for an AR display of an AR wearable device begins at operation **1302** with accessing data indicating a number of gaze fixations per vertical gaze angles. For example, the simulator module **620** accesses the simulation data **622**.

The method **1300** continues at operation **1304** with selecting the tilt angle for the AR display based on a number of the gaze fixations being within a field of view of the AR display, where the tilt angle indicates the field of view of the AR display. For example, the select AR display tilt module **632** selects the vertical tilt **618** of the AR display **616** based on the applications **604** and/or tasks **606** that the user **608** will or is using.

The method **1300** continues at operation **1306** with setting the tilt angle for the AR display. For example, the select AR display tilt module **632** may cause the vertical tilt **618** to change or the AR wearable device **602** may be manufactured to have the vertical tilt **618** of the AR display **616** set to the selected vertical tilt **618** angle.

The method **1300** may include one or more additional operations. Operations of method **1300** may be performed in a different order. One or more of the operations of method **1300** may be optional. The method **1300** may be performed by the client device **102**, system **600**, glasses **1200**, another electronic device, or in another way. Portions of the functionality may be performed on a server computer or host computer. For example, glasses **1200** may be coupled to a host client device **102** or application server **114** where one or more of the operations are performed.

Machine Architecture

FIG. **14** is a diagrammatic representation of the machine **1400** within which instructions **1410** (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine **1400** to perform any one or more of the methodologies discussed herein may be executed. For example, the instructions **1410** may cause the machine **1400** to execute any one or more of the methods described herein. The instructions **1410** transform the general, non-programmed machine **1400** into a particular machine **1400** programmed to carry out the described and

illustrated functions in the manner described. The machine **1400** may operate as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **1400** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **1400** may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smartphone, a mobile device, a wearable device (e.g., a smartwatch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions **1410**, sequentially or otherwise, that specify actions to be taken by the machine **1400**. Further, while only a single machine **1400** is illustrated, the term “machine” shall also be taken to include a collection of machines that individually or jointly execute the instructions **1410** to perform any one or more of the methodologies discussed herein. The machine **1400**, for example, may comprise the client device **102** or any one of a number of server devices forming part of the messaging server system **108**. In some examples, the machine **1400** may also comprise both client and server systems, with certain operations of a particular method or algorithm being performed on the server-side and with certain operations of the particular method or algorithm being performed on the client-side.

The machine **1400** may include processors **1404**, memory **1406**, and input/output I/O components **1402**, which may be configured to communicate with each other via a bus **1440**. In an example, the processors **1404** (e.g., a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) Processor, a Complex Instruction Set Computing (CISC) Processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Radio-Frequency Integrated Circuit (RFIC), another processor, or any suitable combination thereof) may include, for example, a processor **1408** and a processor **1412** that execute the instructions **1410**. The term “processor” is intended to include multi-core processors that may comprise two or more independent processors (sometimes referred to as “cores”) that may execute instructions contemporaneously. Although FIG. **14** shows multiple processors **1404**, the machine **1400** may include a single processor with a single-core, a single processor with multiple cores (e.g., a multi-core processor), multiple processors with a single core, multiple processors with multiples cores, or any combination thereof.

The memory **1406** includes a main memory **1414**, a static memory **1416**, and a storage unit **1418**, both accessible to the processors **1404** via the bus **1440**. The main memory **1406**, the static memory **1416**, and storage unit **1418** store the instructions **1410** embodying any one or more of the methodologies or functions described herein. The instructions **1410** may also reside, completely or partially, within the main memory **1414**, within the static memory **1416**, within machine-readable medium **1420** within the storage unit **1418**, within at least one of the processors **1404** (e.g., within the Processor’s cache memory), or any suitable combination thereof, during execution thereof by the machine **1400**.

The I/O components **1402** may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components **1402**

that are included in a particular machine will depend on the type of machine. For example, portable machines such as mobile phones may include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O components **1402** may include many other components that are not shown in FIG. **14**. In various examples, the I/O components **1402** may include user output components **1426** and user input components **1428**. The user output components **1426** may include visual components (e.g., a display such as a plasma display panel (PDP), a light-emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The user input components **1428** may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or another pointing instrument), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

In further examples, the I/O components **1402** may include biometric components **1430**, motion components **1432**, environmental components **1434**, or position components **1436**, among a wide array of other components. For example, the biometric components **1430** include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye-tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram-based identification), and the like. The motion components **1432** include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope).

The environmental components **1434** include, for example, one or cameras (with still image/photograph and video capabilities), illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometers that detect ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detection concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment.

With respect to cameras, the client device **102** may have a camera system comprising, for example, front cameras on a front surface of the client device **102** and rear cameras on a rear surface of the client device **102**. The front cameras may, for example, be used to capture still images and video of a user of the client device **102** (e.g., “selfies”), which may then be augmented with augmentation data (e.g., filters) described above. The rear cameras may, for example, be used to capture still images and videos in a more traditional camera mode, with these images similarly being augmented with augmentation data. In addition to front and rear cam-

eras, the client device **102** may also include a 360° camera for capturing 360° photographs and videos.

Further, the camera system of a client device **102** may include dual rear cameras (e.g., a primary camera as well as a depth-sensing camera), or even triple, quad or penta rear camera configurations on the front and rear sides of the client device **102**. These multiple cameras systems may include a wide camera, an ultra-wide camera, a telephoto camera, a macro camera and a depth sensor, for example.

The position components **1436** include location sensor components (e.g., a GPS receiver component), altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.

Communication may be implemented using a wide variety of technologies. The I/O components **1402** further include communication components **1438** operable to couple the machine **1400** to a network **1422** or devices **1424** via respective coupling or connections. For example, the communication components **1438** may include a network interface Component or another suitable device to interface with the network **1422**. In further examples, the communication components **1438** may include wired communication components, wireless communication components, cellular communication components, Near Field Communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), Wi-Fi® components, and other communication components to provide communication via other modalities. The devices **1424** may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a USB).

Moreover, the communication components **1438** may detect identifiers or include components operable to detect identifiers. For example, the communication components **1438** may include Radio Frequency Identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Data-glyph, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components **1438**, such as location via Internet Protocol (IP) geolocation, location via Wi-Fi® signal triangulation, location via detecting an NFC beacon signal that may indicate a particular location, and so forth.

The various memories (e.g., main memory **1414**, static memory **1416**, and memory of the processors **1404**) and storage unit **1418** may store one or more sets of instructions and data structures (e.g., software) embodying or used by any one or more of the methodologies or functions described herein. These instructions (e.g., the instructions **1410**), when executed by processors **1404**, cause various operations to implement the disclosed examples.

The instructions **1410** may be transmitted or received over the network **1422**, using a transmission medium, via a network interface device (e.g., a network interface component included in the communication components **1438**) and using any one of several well-known transfer protocols (e.g., hypertext transfer protocol (HTTP)). Similarly, the instructions **1410** may be transmitted or received using a transmission medium via a coupling (e.g., a peer-to-peer coupling) to the devices **1424**.

Software Architecture

FIG. **15** is a block diagram **1500** illustrating a software architecture **1504**, which can be installed on any one or more of the devices described herein. The software architecture **1504** is supported by hardware such as a machine **1502** that includes processors **1520**, memory **1526**, and I/O components **1538**. In this example, the software architecture **1504** can be conceptualized as a stack of layers, where each layer provides a particular functionality. The software architecture **1504** includes layers such as an operating system **1512**, libraries **1510**, frameworks **1508**, and applications **1506**. Operationally, the applications **1506** invoke API calls **1550** through the software stack and receive messages **1552** in response to the API calls **1550**.

The operating system **1512** manages hardware resources and provides common services. The operating system **1512** includes, for example, a kernel **1514**, services **1516**, and drivers **1522**. The kernel **1514** acts as an abstraction layer between the hardware and the other software layers. For example, the kernel **1514** provides memory management, processor management (e.g., scheduling), component management, networking, and security settings, among other functionalities. The services **1516** can provide other common services for the other software layers. The drivers **1522** are responsible for controlling or interfacing with the underlying hardware. For instance, the drivers **1522** can include display drivers, camera drivers, BLUETOOTH® or BLUETOOTH® Low Energy drivers, flash memory drivers, serial communication drivers (e.g., USB drivers), WI-FI® drivers, audio drivers, power management drivers, and so forth.

The libraries **1510** provide a common low-level infrastructure used by the applications **1506**. The libraries **1510** can include system libraries **1518** (e.g., C standard library) that provide functions such as memory allocation functions, string manipulation functions, mathematic functions, and the like. In addition, the libraries **1510** can include API libraries **1524** such as media libraries (e.g., libraries to support presentation and manipulation of various media formats such as Moving Picture Experts Group-4 (MPEG4), Advanced Video Coding (H.264 or AVC), Moving Picture Experts Group Layer-3 (MP3), Advanced Audio Coding (AAC), Adaptive Multi-Rate (AMR) audio codec, Joint Photographic Experts Group (JPEG or JPG), or Portable Network Graphics (PNG)), graphics libraries (e.g., an OpenGL framework used to render in two dimensions (2D) and three dimensions (3D) in a graphic content on a display), database libraries (e.g., SQLite to provide various relational database functions), web libraries (e.g., WebKit to provide web browsing functionality), and the like. The libraries **1510** can also include a wide variety of other libraries **1528** to provide many other APIs to the applications **1506**.

The frameworks **1508** provide a common high-level infrastructure that is used by the applications **1506**. For example, the frameworks **1508** provide various graphical user interface (GUI) functions, high-level resource management, and high-level location services. The frameworks **1508** can provide a broad spectrum of other APIs that can be used by the applications **1506**, some of which may be specific to a particular operating system or platform.

In an example, the applications **1506** may include a home application **1536**, a contacts application **1530**, a browser application **1532**, a reader application **1534**, a location application **1542**, a media application **1544**, a messaging application **1546**, a game application **1548**, and a broad assortment of other applications such as a third-party application **1540**. The applications **1506** are programs that

execute functions defined in the programs. Various programming languages can be employed to create one or more of the applications **1506**, structured in a variety of manners, such as object-oriented programming languages (e.g., Objective-C, Java, or C++) or procedural programming languages (e.g., C or assembly language). In a specific example, the third-party application **1540** (e.g., an application developed using the ANDROID™ or IOS™ software development kit (SDK) by an entity other than the vendor of the particular platform) may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® Phone, or another mobile operating system. In this example, the third-party application **1540** can invoke the API calls **1550** provided by the operating system **1512** to facilitate functionality described herein.

Processing Components

Turning now to FIG. **16**, there is shown a diagrammatic representation of a processing environment **1600**, which includes a processor **1602**, a processor **1606**, and a processor **1608** (e.g., a GPU, CPU, or combination thereof).

The processor **1602** is shown to be coupled to a power source **1604**, and to include (either permanently configured or temporarily instantiated) modules, namely a graphics display component **1610**, a simulation component **1612**, and a select AR display tilt component **1614**. The graphics display component **1610** controls an AR display such as AR display **616**. The simulation component **1612** processes simulation data **622** to generate processed simulation data **630**. The simulation component **1612** is the same or similar as the simulator module **620** described in conjunction with FIG. **6**, in accordance with some embodiments.

The select AR display tilt component **1614** selects the vertical tilt **618** of the AR display **616**. For example, the select AR display tilt module **632** selects the vertical tilt **618** angle of the AR display **616** of the AR wearable device **602**. As illustrated, the processor **1602** is communicatively coupled to both the processor **1606** and the processor **1608**.

Glossary

Certain examples are described herein as including logic or a number of components, modules, or mechanisms. Modules may constitute either software modules (e.g., code embodied on a machine-readable medium or in a transmission signal) or hardware modules. A “hardware module” is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various example examples, one or more computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or one or more hardware modules of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware module that operates to perform certain operations as described herein.

“Carrier signal” refers to any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the machine, and includes digital or analog communications signals or other intangible media to facilitate communication of such instructions. Instructions may be transmitted or received over a network using a transmission medium via a network interface device.

“Client device” refers to any machine that interfaces to a communications network to obtain resources from one or more server systems or other client devices. A client device may be, but is not limited to, a mobile phone, an AR glasses, a VR glasses, an AR wearable device, a desktop computer,

a laptop, a portable digital assistants (PDAs), smartphones, tablets, ultrabooks, netbooks, laptops, multi-processor systems, microprocessor-based or programmable consumer electronics, game consoles, set-top boxes, or any other communication device that a user may use to access a network.

“Communication network” refers to one or more portions of a network that may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of network, or a combination of two or more such networks. For example, a network or a portion of a network may include a wireless or cellular network and the coupling may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or other types of cellular or wireless coupling. In this example, the coupling may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1xRTT), Evolution-Data Optimized (EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP) including 3G, fourth generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long Term Evolution (LTE) standard, others defined by various standard-setting organizations, other long-range protocols, or other data transfer technology.

“Component” refers to a device, physical entity, or logic having boundaries defined by function or subroutine calls, branch points, APIs, or other technologies that provide for the partitioning or modularization of particular processing or control functions. Components may be combined via their interfaces with other components to carry out a machine process. A component may be a packaged functional hardware unit designed for use with other components and a part of a program that usually performs a particular function of related functions. Components may constitute either software components (e.g., code embodied on a machine-readable medium) or hardware components. A “hardware component” is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various examples, one or more computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or one or more hardware components of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware component that operates to perform certain operations as described herein. A hardware component may also be implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware component may include dedicated circuitry or logic that is permanently configured to perform certain operations. A hardware component may be a special-purpose processor, such as a field-programmable gate array (FPGA) or an application specific integrated circuit (ASIC). A hardware component may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware component may

include software executed by a general-purpose processor or other programmable processor. Once configured by such software, hardware components become specific machines (or specific components of a machine) uniquely tailored to perform the configured functions and are no longer general-purpose processors. It will be appreciated that the decision to implement a hardware component mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software), may be driven by cost and time considerations. Accordingly, the phrase “hardware component” (or “hardware-implemented component”) should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. Considering examples in which hardware components are temporarily configured (e.g., programmed), each of the hardware components need not be configured or instantiated at any one instance in time. For example, where a hardware component comprises, a general-purpose processor configured by software to become a special-purpose processor, the general-purpose processor may be configured as respectively different special-purpose processors (e.g., comprising different hardware components) at different times. Software accordingly configures a particular processor or processors, for example, to constitute a particular hardware component at one instance of time and to constitute a different hardware component at a different instance of time. Hardware components can provide information to, and receive information from, other hardware components. Accordingly, the described hardware components may be regarded as being communicatively coupled. Where multiple hardware components exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) between or among two or more of the hardware components. In examples in which multiple hardware components are configured or instantiated at different times, communications between such hardware components may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware components have access. For example, one hardware component may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware component may then, at a later time, access the memory device to retrieve and process the stored output. Hardware components may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information). The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented components that operate to perform one or more operations or functions described herein. As used herein, “processor-implemented component” refers to a hardware component implemented using one or more processors. Similarly, the methods described herein may be at least partially processor-implemented, with a particular processor or processors being an example of hardware. For example, at least some of the operations of a method may be performed by one or more processors or processor-implemented components. Moreover, the one or more processors may also operate to support performance of the relevant operations in a “cloud computing” environment

or as a “software as a service” (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines including processors), with these operations being accessible via a network (e.g., the Internet) and via one or more appropriate interfaces (e.g., an API). The performance of certain of the operations may be distributed among the processors, not only residing within a single machine, but deployed across a number of machines. In some examples, the processors or processor-implemented components may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other examples, the processors or processor-implemented components may be distributed across a number of geographic locations.

“Computer-readable storage medium” refers to both machine-storage media and transmission media. Thus, the terms include both storage devices/media and carrier waves/modulated data signals. The terms “machine-readable medium,” “computer-readable medium” and “device-readable medium” mean the same thing and may be used interchangeably in this disclosure.

“Ephemeral message” refers to a message that is accessible for a time-limited duration. An ephemeral message may be a text, an image, a video and the like. The access time for the ephemeral message may be set by the message sender. Alternatively, the access time may be a default setting or a setting specified by the recipient. Regardless of the setting technique, the message is transitory.

“Machine storage medium” refers to a single or multiple storage devices and media (e.g., a centralized or distributed database, and associated caches and servers) that store executable instructions, routines and data. The term shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media, including memory internal or external to processors. Specific examples of machine-storage media, computer-storage media and device-storage media include non-volatile memory, including by way of example semiconductor memory devices, e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), FPGA, and flash memory devices; magnetic disks such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks. The terms “machine-storage medium,” “device-storage medium,” “computer-storage medium” mean the same thing and may be used interchangeably in this disclosure. The terms “machine-storage media,” “computer-storage media,” and “device-storage media” specifically exclude carrier waves, modulated data signals, and other such media, at least some of which are covered under the term “signal medium.”

“Non-transitory computer-readable storage medium” refers to a tangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine.

“Signal medium” refers to any intangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine and includes digital or analog communications signals or other intangible media to facilitate communication of software or data. The term “signal medium” shall be taken to include any form of a modulated data signal, carrier wave, and so forth. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. The terms “transmission medium” and “signal medium” mean the same thing and may be used interchangeably in this disclosure.

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What is claimed is:

1. A method for setting a tilt angle for an augmented reality (AR) display of an augmented reality (AR) wearable device, the method comprising:
 - accessing data indicating a number of gaze fixations per vertical gaze angles, the number of gaze fixations being associated with a task of an application a user of the AR wearable device is performing;
 - selecting the tilt angle for the AR display based on the number of the gaze fixations being within a field of view of the AR display, wherein the tilt angle indicates the field of view of the AR display; and
 - setting the tilt angle for the AR display.
2. The method of claim 1 wherein the vertical gaze angles are from +30 degrees to -40 degrees.
3. The method of claim 1 further comprising:
 - collecting the data for a plurality of applications by monitoring the gaze fixations of users of the plurality of applications while the users perform tasks using the plurality of applications.
4. The method of claim 3 wherein the plurality of applications is a first plurality of applications and wherein the method further comprises:
 - selecting the first plurality of applications from a second plurality of applications based on a probability the first plurality of applications will be used by the user of the AR wearable device.
5. The method of claim 1 wherein the tilt angle is selected based on the number of gaze fixations being a largest number within the field of view of the AR display for the tilt angle.
6. The method of claim 1 wherein the AR wearable device has a fixed field of view that is larger than the field of view of the AR display.
7. The method of claim 6 wherein the fixed field of view is of a real-world scene, wherein a user of the AR wearable device sees the fixed field of view through optical elements of the AR wearable device.
8. The method of claim 7 wherein the tilt angle is a first tilt angle and the optical elements have a second tilt angle, and wherein the first tilt angle is measured based on the second tilt angle.
9. The method of claim 1 further comprising:
 - manufacturing the AR wearable device with the tilt angle for the AR display.
10. The method of claim 1 wherein setting the tilt angle for the AR display further comprises:
 - causing an actuator to change a current tilt angle for the AR display to the tilt angle for the AR display.
11. The method of claim 1 further comprising:
 - adjusting the number of gaze fixations per vertical gaze angles based on a noise distribution, wherein the noise distribution is an angle the AR wearable device is worn by users of the AR wearable device.
12. The method of claim 1 wherein the selecting the tilt angle is further based on a gaze range of the task and the application.

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13. An augmented reality (AR) wearable device comprising:
 - at least one processor; and
 - a memory storing instructions that, when executed by the at least one processor, configure the AR wearable device to perform operations comprising:
 - accessing data indicating a number of gaze fixations per vertical gaze angles, the number of gaze fixations being associated with a task of an application a user of the AR wearable device is performing;
 - selecting a tilt angle for an AR display of the AR wearable device based on the number of the gaze fixations being within a field of view of the AR display, wherein the tilt angle indicates the field of view of the AR display; and
 - setting the tilt angle for the AR display.
14. The AR wearable device of claim 13 wherein the vertical gaze angles are from +30 degrees to -40 degrees.
15. The AR wearable device of claim 13 wherein the operations further comprise:
 - collecting the data for a plurality of applications by monitoring the gaze fixations of users of the plurality of applications while the users perform tasks using the plurality of applications.
16. The AR wearable device of claim 13 wherein setting the tilt angle for the AR display further comprises:
 - causing an actuator to change a current tilt angle for the AR display to the tilt angle for the AR display.
17. A non-transitory computer-readable storage medium, the computer-readable storage medium including instructions that when executed by an augmented reality (AR) wearable device, cause the AR wearable device to perform operations comprising:
 - accessing data indicating a number of gaze fixations per vertical gaze angles, the number of gaze fixations being associated with a task of an application a user of the AR wearable device is performing;
 - selecting a tilt angle for an AR display of the AR wearable device based on the number of the gaze fixations being within a field of view of the AR display, wherein the tilt angle indicates the field of view of the AR display; and
 - setting the tilt angle for the AR display.
18. The non-transitory computer-readable storage medium of claim 17 wherein the vertical gaze angles are from +30 degrees to -40 degrees.
19. The non-transitory computer-readable storage medium of claim 17 wherein the operations further comprise:
 - collecting the data for a plurality of applications by monitoring the gaze fixations of users of the plurality of applications while the users perform tasks using the plurality of applications.
20. The non-transitory computer-readable storage medium of claim 17 wherein setting the tilt angle for the AR display further comprises:
 - causing an actuator to change a current tilt angle for the AR display to the tilt angle for the AR display.

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